

THE MAHATMA GANDHI UNIVERSITY
UNDER GRADUATE VOCATIONAL
PROGRAMMES (HONOURS)
SYLLABUS

MGU-B.VOC. (Honours)

(2025 Admission Onwards)



Faculty: Technology and Applied Sciences

Expert Committee: Creative Knowledge Based Careers

Programme: B.Voc. (Honours) Sound Engineering

Mahatma Gandhi University
Priyadarshini Hills
Kottayam – 686560, Kerala, India

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MGU-B.VOC. (HONOURS)

Syllabus

Preface

The rapid evolution of technology and the growing demand for skilled professionals in the creative and entertainment industries have transformed the landscape of higher education. Recognizing the need for a dynamic and application-oriented curriculum, Mahatma Gandhi University has introduced the Bachelor of Vocation (B.Voc.) Degree Programme in Sound Engineering, an academic model designed to bridge the gap between industry expectations and educational outcomes.

The curriculum encompasses a comprehensive range of subjects—from science of sound, audio electronics, DAW fundamentals, and live sound to media management, entrepreneurship, dubbing, surround sound production, and studio acoustics. These modules are enriched with internships, industry exposure, project work, and personality development, ensuring that students not only master sound engineering techniques but also develop soft skills, professional ethics, and entrepreneurial competencies.

In addition to fostering technical proficiency, this curriculum emphasizes cultural relevance, regional diversity, and sustainable practices in sound and media. It equips learners with the knowledge to navigate both traditional and modern audio technologies, while also encouraging innovation in areas such as radio production, music programming, mixing, mastering, and sound for visual media.

Designed to keep pace with the changing media environment and the creative economy, this curriculum aims to empower students to pursue careers as sound engineers, audio producers, music directors, media technologists, and entrepreneurs not just in India, but globally.

We believe that this curriculum will serve as a catalyst for producing technically sound, socially responsible, and industry ready graduates who can contribute meaningfully to the world of sound and media.

Expert Committee & External Experts

1. Dr. Seethal S Nair (Convener), Assistant Professor of English, NSS Hindu College, Changanacherry
2. Dr. Shynu Abraham, Assistant Professor of Malayalam, Catholicate College, Pathanamthitta
3. Dr. Divya S Nair, Assistant Professor of English NSS Hindu College, Changanacherry
4. Shreeja Narayanan, Assistant Professor of English, Sree Sankara College, Kalady
5. Archana Sankar, Assistant Professor(B.Voc), St. Paul's College, Kalamassery
6. Nandu R, B.Voc Nodal Officer, Al Ameen College, Edathala, Aluva
7. Annie Linda Hari, Assistant Professor(B.Voc) Mar Thoma College for Women, Perumbavoor
8. Saru Tresa Sunny, Assistant Professor(B.Voc), Alphonsa College, Pala
9. Manu V. G B.Voc Nodal Officer, Pavanatma College, Murickassery
10. Zeba Nazar, Assistant Professor(B.Voc), St. Theresa's College, Ernakulam
11. Bibu V V, Assistant Professor of Journalism, St. Xavier's College, Vaikom
12. Dr. Lekha Sreenivas, Associate Professor of Fashion Designing, St. Theresa's College, Ernakulam
13. Dr. Vinitha Paulose, Assistant Professor of Fashion Designing, St. Theresa's College, Ernakulam
14. Elena Sajan Assistant Professor of English, St. Thomas College, Ranni
15. Dr. Ranjini Radhakrishnan, Assistant Professor of Physics, K E College, Mannanam

External Experts

1. Arun Rama Varma, Sound Designer, Sound Editor, Indian Film Industry
2. Arun Babu, Sound Engineer, Writer, Indian Film Industry

Syllabus Index

Name of the Major Subject: **Sound Engineering**

SEMESTER: 1

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG1SDCSEG100	Science of Sound	SDC	4	4	4	0	0
MG1SDCSEG101	Audio Electronics	SDC	4	5	3	2	0
MG1SDCSEG102	Microphone Design and Application	SDC	4	5	3	2	0
MG1MDCSEG100	Psychoacoustics	MDC	3	3	3	0	0
MG1SDCSEG103	On the Job Training	SDC	2	5	0	0	5

L — Lecture, P — Practical/Practicum , O — On the Job Training

SEMESTER: 2

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG2SDCSEG100	Analog and Digital Audio	SDC	4	4	4	0	0
MG2SDCSEG101	Audio Signal Processing	SDC	4	5	3	2	0
MG2SDCSEG102	DAW Fundamentals	SDC	4	5	3	2	0
MG2MDCSEG100	Music Composition	MDC	3	4	2	2	0
MG2SDCSEG103	On the Job Training	SDC	2	5	0	0	5

L — Lecture, P — Practical/Practicum , O — On the Job Training

SEMESTER: 3

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG3SDCSEG200	Music Architecture	SDC	4	4	4	0	0
MG3SDCSEG201	Sampling and Synthesis	SDC	4	5	3	2	0
MG3SDCSEG202	Live Sound Fundamentals	SDC	4	5	3	2	0
MG3MDCSEG200	Sound, Society and Technology	MDC	3	3	3	0	0
MG3SDCSEG203	On the Job Training	SDC	2	5	0	0	5

L — Lecture, P — Practical/Practicum , O — On the Job Training

SEMESTER: 4

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG4SDCSEG200	Media Management	SDC	4	4	4	0	0
MG4SDCSEG201	MIDI and Music Programming	SDC	4	5	3	2	0
MG4SDCSEG202	Sound Story	SDC	4	5	3	2	0
MG4SECSEG200	Sound for Broadcasting and Live Events	SEC	3	3	3	0	0
MG4VACSEG200	Electrical Aspects in Sound Engineering	VAC	3	3	3	0	0
MG4INTSEG200	Internship	INT	2	20 days on Sem Break			

L — Lecture, P — Practical/Practicum , O — On the Job Training

SEMESTER: 5

Course Code	Title of the Course	Type of the Course		Credit	Hours/ week	Hour Distribution /week		
						L	P	O
MG5SDCSEG300	New Media	SDC		4	4	4	0	0
MG5SDESEG300	Studio Acoustics	SDE	Select any one from SDE	4	5	3	2	0
MG5SDESEG301	Stereo Amplifier Assembling and Testing	SDE		4	5	3	2	0
MG5SDCSEG301	Multitrack Audio Production	SDC		4	5	3	2	0
MG5SECSEG300	Art and Technology of Mixing and Mastering	SEC		3	4	2	2	0
MG5VACSEG300	Sonic Awareness and Public Responsibility	VAC		3	3	3	0	0

L — Lecture, P — Practical/Practicum , O — On the Job Training

SEMESTER: 6

Course Code	Title of the Course	Type of the Course		Credit	Hours/ week	Hour Distribution /week		
						L	P	O
MG6SDESEG300	Dubbing and Sync Sound	SDE	Select any one from SDE	4	5	3	2	0
MG6SDESEG301	Audio Remastering Techniques	SDE		4	5	3	2	0
MG6SDCSEG300	Surround Sound Production	SDC		4	5	3	2	0
MG6SECSEG300	Principle and Techniques of Radio Production	SEC		3	3	3	0	0
MG6VACSEG300	Art of Critical Listening	VAC		3	3	3	0	0
MG6PRJSEG300	Project	PRJ		4	8	0	8	0

L — Lecture, P — Practical/Practicum , O — On the Job Training

B.VOC HONOURS

SEMESTERS: 7 & 8

Course Code	Title of the Course	Type of the Course	Credit	Number of days	Credits		
					L	P	O
MG7APPSEG400	Apprenticeship	APP	28	280 days	0	28	0

L — Lecture, P — Practical/Practicum , O — On the Job Training

B.VOC HONOURS WITH RESEARCH

SEMESTERS: 7 & 8

Course Code	Title of the Course	Type of the Course	Credit	Number of days	Credits		
					L	P	O
MG7RINSEG400	Research Internship	RIN	20	200 days	0	20	0
	Online	SDC	4	NA	4	0	0
	Online	SDC	4	NA	4	0	0

L — Lecture, P — Practical/Practicum , O — On the Job Training

MGU-B.VOC. (HONOURS)

Syllabus

Job Roles and Qualification Packs for Certificate, Diploma, Bachelor's, and Honours Degrees.

JOB ROLES	NHEQF LEVEL	QPS ALIGNED	SECTOR SKILL	YEAR
Sound Designer	Level 5	MES/N3401 MES/N3402 MES/N3403 MES/N0104 DGT/VSQ/N0101 OJT	MESC	Second



MGU-B.VOC. (HONOURS)

Syllabus



SEMESTER 1

MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering			
Programme	MGU-B.Voc. (Honours) Sound Engineering			
Course Name	Science of Sound			
Type of Course	SDC			
Course Code	MG1SDCSEG100			
Course Level	100			
Course Summary	This course introduces students to the physics of sound, exploring how sound is created, transmitted, and heard. Key topics include sound waves, frequency, amplitude, pitch, resonance, and the behaviour of sound in different environments. Students will also examine practical applications such as musical acoustics, hearing, and sound technology. Through hands-on experiments and real-world examples, learners gain a deeper understanding of how sound shapes our world.			
Semester	1	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
	Lecture class, Hands-on Practicum sessions	4	0	0
Pre- requisites, if any	Nil			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Explain the fundamental wave properties of sound, including waveform characteristics, propagation through different media, and physical behaviors like reflection, refraction, and diffraction	U	1,2,3,4,6,7,8,9,10
2	Understand audio measurements and sound levels using concepts like decibels, SPL, metering types, and understand parameters such as dynamic range, headroom, and signal-to-noise ratio.	U	1,2,3,4,5,6,7,8,9,10
3	Describe the structure and function of the human ear, and explain how it perceives sound in terms of pitch, loudness, frequency sensitivity, and protective mechanisms.	U	1,2,3,4,5,6,7,8,9,10
4	Apply the acoustic principles to understand human sound perception, including binaural hearing, localization, masking, the Doppler effect, and auditory phenomena in complex environments.	A	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	3	2	1	0	1	1	1	1	2
CO2	2	3	2	1	1	1	1	1	1	2
CO3	2	3	2	2	1	1	1	1	2	2
CO4	2	3	2	1	2	1	1	1	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Wave Nature and Characteristics		15	
	1.1	Wave theory – Sine wave, Transverse and Longitudinal wave, Sound as a pressure wave, Propagation of sound wave through medium, Waveform characteristics – Amplitude, Frequency, Wavelength, Time-period, Velocity, Phase.	5	1
	1.2	Sound and density, Physical characteristics of sound – reflection, absorption, refraction diffraction, diffusion	5	1
	1.3	Simple and complex wave, Fundamental frequency, Harmonics and Overtones, Partial, Octave, Timbre; Different waveform types – Sine, Square, Triangle, Saw-tooth; Sound Envelope – ADSR	5	1
2	Audio Metering		15	
	2.1	Audio Measurement Basics: Learn the difference between Peak and RMS values, and how sound is measured using linear and logarithmic scales. Understand the concept of decibels (dB), sound pressure levels (SPL), and how sound pressure is converted into dB using specific equations and the inverse-square law.	3	2
	2.2	SPL, Metering, and Standards: Explore SPL measurements, from the threshold of hearing to the threshold of pain. Understand metering tools like VU and PPM meters, and units such as dBm, dBu, dBV, dBA, dBFS, and LUFS. Learn about weighting networks and industry-standard levels for audio equipment interconnection.	7	2

	2.3	Audio Signal Quality: Study key performance indicators including dynamic range, signal-to-noise (S/N) ratio, and headroom. Discover the use of pink and white noise in testing and calibration to ensure high-quality audio output and accurate system performance	5	2
3	Human Ear		15	
	3.1	Structure of the Human Ear: Understand the anatomy of the ear, divided into three main parts—outer ear, middle ear, and inner ear—each playing a crucial role in capturing and processing sound. Learn how the ear perceives pitch and processes sound through critical bands. Explore the concept of dynamic range of hearing and how humans detect variations in loudness and frequency.	7	3
	3.2	Loudness and Ear Protection: Study the Equal Loudness Contour (Fletcher-Munson curve) which shows how the ear perceives loudness at different frequencies, and understand the ear's built-in protective mechanisms against loud or damaging sounds.	8	3
4	Auditory Perception		15	
	4.1	Sound Perception and Localization: Explore how the brain processes correlated and uncorrelated sounds, and how binaural localization helps us determine the direction of sound. Learn about the Haas Effect, where slight delays affect perceived location and clarity.	6	4
	4.2	Auditory Phenomena: Understand key concepts such as masking (when one sound hides another), beats (interference between close frequencies), and the Doppler	6	4

		Effect (perceived pitch change due to motion of the sound source).		
	4.3	Complex Listening Environments: Discover the Cocktail Party Effect, which describes the brain's ability to focus on a single sound source in a noisy environment, demonstrating the complexity of human	3	4
5	5.1	Teacher Specific Content		

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Theory Classes • Demonstrations • Assessments: Regular quizzes, assignments, lab work, and practical exams to check understanding and skills.
Assessment Types	MODE OF ASSESSMENT Mode of Assessment A. Continuous Comprehensive Assessment (CCA) Theory Total Marks: 30 FA: Assignments, Oral Presentations, Viva Voice SA: Written test

	B. End Semester Evaluation (ESE) Theory Total Marks: 70, Duration: 2 hrs <table border="1" data-bbox="435 443 1506 712"> <thead> <tr> <th>Components</th><th>Marks</th></tr> </thead> <tbody> <tr> <td>Very short answer</td><td>10 out of 12 – 10 x 2 = 20 marks</td></tr> <tr> <td>Short answer</td><td>6 out of 9 – 6 x 5 = 30 marks</td></tr> <tr> <td>Essay</td><td>2 out of 3 – 10 x 2 = 20 marks</td></tr> </tbody> </table>	Components	Marks	Very short answer	10 out of 12 – 10 x 2 = 20 marks	Short answer	6 out of 9 – 6 x 5 = 30 marks	Essay	2 out of 3 – 10 x 2 = 20 marks
Components	Marks								
Very short answer	10 out of 12 – 10 x 2 = 20 marks								
Short answer	6 out of 9 – 6 x 5 = 30 marks								
Essay	2 out of 3 – 10 x 2 = 20 marks								

References

1. Huber, David Miles, and Robert E. Runstein. *Modern Recording Techniques*. 9th ed. New York: Focal Press, 2017.
2. Rossing, Thomas D., Richard F. Moore, and Paul A. Wheeler. *The Science of Sound*. 3rd ed. San Francisco: Addison-Wesley, 2002.
3. Kinsler, Lawrence E., Austin R. Frey, Alan B. Coppens, and James V. Sanders. *Fundamentals of Acoustics*. 4th ed. Hoboken, NJ: Wiley, 2000.
4. Howard, David M., and Jamie A. S. Angus. *Acoustics and Psychoacoustics*. 5th ed. New York: Focal Press, 2017.



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering			
Programme	MGU-B.Voc. (Honours) Sound Engineering			
Course Name	Audio Electronics			
Type of Course	SDC			
Course Code	MG1SDCSEG101			
Course Level	100			
Course Summary	This course brings together the basics of electronics and their use in sound engineering. It starts with simple ideas like atoms and electricity, then moves on to important components like resistors, capacitors, transformers, and semiconductors. Students also learn about different cables and connections used in audio systems. Finally, the course gives hands-on practice in soldering and using a multi-meter, helping students build both knowledge and practical skills for working with audio equipment.			
Semester	1	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
	Lecture class, Hands-on Practicum sessions	3	1	0
Pre- requisites, if any				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Remember the fundamental principles of atomic structure, electricity, magnetism, and basic electronic concepts including current, voltage, resistance, and circuit configurations.	K	1,2,3,4,6, 7,8,9,10

2	Understand the functions of passive and active electronic components such as resistors, capacitors, inductors, transformers, semiconductors, diodes, transistors, ICs, and operational amplifiers.	U	1,2,3,4,5,6,7,8,9,10
3	Apply transmission techniques, differentiate between various audio cables, connectors, and understand impedance, balanced/unbalanced transmission, and frequency response in audio systems.	A	1,2,3,4,5,6,7,8,9,10
4	Demonstrate practical skills in soldering, cable assembly, and using multimeters to test and troubleshoot audio cables and electronic components	A	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	3	2	1	0	1	1	1	1	2
CO2	2	3	2	1	1	1	1	1	1	2
CO3	2	3	2	2	1	1	1	1	2	2
CO4	2	3	2	1	2	1	1	1	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Basic Electronics		15	
	1.1	Atomic Structure – Electron, Proton, Neutron; Atom Model, Free Electrons	3	1
	1.2	Conductor, Insulator, Semiconductor, Current, Voltage, Batteries, Resistance, Ohms Law, Power, Series and Parallel Connections	7	1

	1.3	Electricity and Magnetism, Faraday's Law, Electromagnetic Induction, Mutual Induction AC current, DC current	5	1
2	Active & Passive Components		15	
	2.1	Resistor – Introduction, Basic Principles, Resistance of a resistor, Resistor colour codes, Series and Parallel connection, Variable Resistor Capacitor – Introduction, Basic Principles, capacitance, types of caps, series and parallel connection Inductor – Introduction, Inductance measurement, Coils	5	2
	2.2	Transformers – Basics, Step-up, Step-down Vacuum Tubes – Basics	3	2
	2.3	Semiconductors – Introduction, p- type and n-type, Diodes – basics, biasing, diode rectifier; LED's, Transistors – Transistor Types, functions; IC's, Op-Amp's Interpreting circuit diagrams – Basics, Common symbols	7	2
3	Transmission techniques and General Audio Cable types		15	
	3.1	Cable and its characteristics - Balanced & Unbalanced Transmission, DI Box, Impedance, Frequency response	5	3
	3.2	General Audio Cable types - Twisted pair cable, snake cable, shielded cable, Mono/Stereo cable, co-axial cables, XLR male cable XLR Female cable, XLR Board connector, TRS male cable, RCA male, RCA female, RCA Cable, DIN cable, USB cable, DB25, BNC, LAN Cable, Concept of OFC and its applications – ADAT cable	10	3

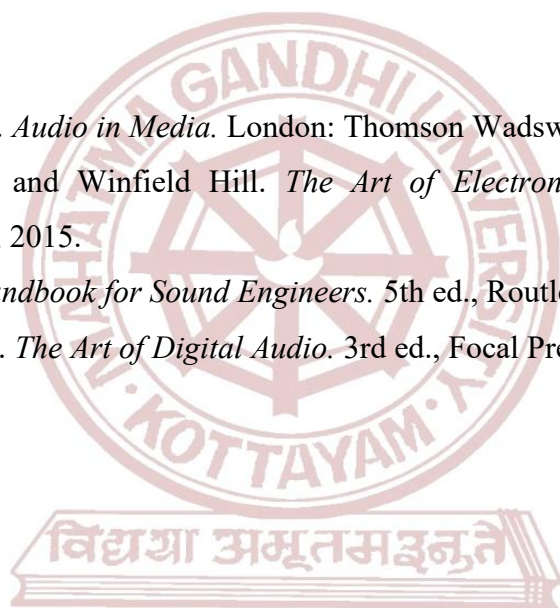
4	Practical Session		30	
	4.1	Soldering – Basics, necessary components – Soldering Iron, Solder lead, Flux, Solder sucker, Soldering Tips & Techniques	10	4
	4.2	How to solder – XLR Cable, TRS Cable, TS Cable (Projects)	10	4
	4.3	Multimeter – Basics, Uses, Controls, setting up the multimeter, taking voltage, current, resistance measurements, testing various components, testing wires and cables for continuity	10	4
5	5.1	Teacher Specific Content		

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Theory Classes: Easy-to-understand lectures with diagrams, videos, and real-life examples to explain basic electronics, digital audio, and sound recording concepts. • Demonstrations: Live demonstrations of electronic components, soldering, cable connections, and multi-meter usage. • Practical Sessions: Hands-on practice with circuit building, cable soldering, component testing, and using audio equipment. • Assessments: Regular quizzes, assignments, lab work, and practical exams to check understanding and skills.
Assessment Types	MODE OF ASSESSMENT Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) Practical Total Marks: 30 FA: Viva Voice, Oral Presentations SA: Practical test

	B. End Semester Evaluation (ESE) Practical Total Marks: 70, Duration: 3 hrs Lab Exam: 60 marks Viva Voice: 10 marks
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References

1. Alten, Stanley R. *Audio in Media*. London: Thomson Wadsworth, 2004.
2. Horowitz, Paul, and Winfield Hill. *The Art of Electronics*. 3rd ed., Cambridge University Press, 2015.
3. Ballou, Glen. *Handbook for Sound Engineers*. 5th ed., Routledge, 2015.
4. Watkinson, John. *The Art of Digital Audio*. 3rd ed., Focal Press, 2001.



MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering			
Programme	MGU-B.Voc. (Honours) Sound Engineering			
Course Name	Microphone Design and Application			
Type of Course	SDC			
Course Code	MG1SDCSEG102			
Course Level	100			
Course Summary	To introduce the design, working principles, and types of microphones, develop an understanding of their characteristics and specifications, explore various techniques for different recording scenarios, and provide hands-on training in microphone selection, placement, and recording practices.			
Semester	1	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
	Lecture class, Hands-on Practicum sessions	3	1	0
Pre- requisites, if any	Nil			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Explain the working principles and designs of various microphone types.	U	1,2,3,4, 6,8,9,10
2	Understand microphone characteristics such as polar patterns, frequency response, and transient response.	U	1,2,3,4, 6,8,9,10

3	Apply various microphone techniques and placement methods for diverse instruments and recording situations,	A	1,2,3,4,5,6,7,8,9,10
4	Select and handle microphones appropriately based on brand, application, and recording environment	A	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	3	2	1	0	1	0	1	1	2
CO2	2	3	2	1	0	1	0	1	1	2
CO3	2	3	3	2	1	2	1	1	2	3
CO4	2	2	2	2	1	1	1	2	2	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Microphone Design and Working Principle		15	
	1.1	Introduction to microphone transducers. Types of microphones	3	1
	1.2	Detailed design and working mechanism of each microphone type	7	1
	1.3	Applications and advantages/disadvantages of each design	5	1
2	Microphone Characteristics & Preamps		15	
	2.1	Directional Response: Polar patterns, omnidirectional, cardioid, super cardioid, figure-8, shotgun	5	2
	2.2	Frequency Response & Transient Response	3	2

	2.3	Demonstrations of various polar patterns, Output characteristics and sensitivity.	7	2
3	Microphone Techniques and Applications		15	
	3.1	Mono miking techniques. Stereo miking techniques, AB, MS Pair, ORTF, Binaural, Blumlein, Decca Tree.	5	3
	3.2	Surround microphone setups and ambisonics microphones. Recording challenges: Proximity effect, popping, off-axis pickup. Microphone placement techniques for Vocals	10	3
4	Practical Session		30	
	4.1	Brand-specific microphone analysis: Shure, Telefunken, AKG, Neumann, Rode, Warm Audio, USB microphones.	10	4
	4.2	Microphone shootouts and listening tests. Comparative recording analysis and report submission.	10	4
	4.3	Practical exercises: Setting up microphones for various instruments. Recording sessions with multiple microphone techniques. Troubleshooting microphone issues in recording.	10	4
5	5.1	Teacher Specific Content		

Teaching and Learning Approach	Classroom Procedure (Mode of transaction)
	• Interactive Lectures & Visual Aids Concepts of microphone types, design, and characteristics are taught through classroom lectures, diagrams, videos, and hardware demonstrations. • Hands-On Equipment Demonstration Students engage with real microphones, preamps, and audio interfaces

	to explore polar patterns, frequency response, and placement techniques. • Practical Lab Sessions Regular lab exercises are conducted for mic setup, placement with different instruments, and testing various brands under real-world conditions. • Project-Based & Peer Learning Students apply knowledge in small group projects and assignments involving mic selection, recording tasks, and presentations to reinforce decision-making and real-time problem-solving.
Assessment Types	MODE OF ASSESSMENT Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) Practical Total Marks: 30 FA: Viva Voice, Assignments, Oral Presentation SA: Practical tests
	B. End Semester Evaluation (ESE) Practical Total Marks: 70 , Duration: 3 hrs Lab Exam: 60 marks Viva: 10 marks

Syllabus

References

1. Huber, David Miles. *Modern Recording Techniques*. 9th ed. New York: Focal Press, 2017.
2. Bartlett, Bruce, and Jenny Bartlett. *Practical Recording Techniques*. 5th ed. New York: Focal Press, 2009.
3. Rumsey, Francis, and Tim McCormick. *Sound and Recording: Applications and Theory*. 7th ed. New York: Focal Press, 2014.
4. Owsinski, Bobby. *The Recording Engineer's Handbook*. 4th ed. Milwaukee, WI: Hal Leonard Corporation, 2017.



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering			
Programme	MGU-B.Voc. (Honours) Sound Engineering			
Course Name	Psychoacoustics			
Type of Course	MDC			
Course Code	MG1MDCSEG100			
Course Level	100			
Course Summary	The course aims to explore how humans perceive and process sound, understand the physiological and psychological aspects of hearing, study the cognitive and emotional dimensions of auditory experiences, and relate psychoacoustics to practical applications in sound engineering, design, healthcare, and media.			
Semester	1	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
	Lecture class, Hands-on Practicum sessions	3	0	0
Pre- requisites, if any	Nil			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the structure and functioning of the human auditory system	U	1,2,3,4,6,7, 8,9,10
2	Explain key mechanisms involved in pitch, loudness, timbre perception, spatial hearing, and masking.	U	1,2,3,4,6,7, 8,9,10

3	Understand the influence of sound on cognition, emotion, and memory.	U	1,2,3,4,5,6,7,8,9,10
4	Apply psychoacoustic principles in areas such as audio compression, sound design, hearing aids, and noise control.	A	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	3	2	1	0	1	1	1	1	2
CO2	3	3	2	2	0	1	1	1	1	2
CO3	3	2	2	2	1	2	2	2	2	3
CO4	2	3	3	2	2	2	2	2	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Physiology of Hearing		15	
	1.1	Introduction to the auditory system. Anatomy and functions of the outer, middle, and inner ear. Cochlear function and the process of signal transduction. Types, causes, and effects of hearing loss.	7	1
	1.2	Perception of pitch: frequency and brain interpretation. Loudness perception and sound intensity. Timbre: harmonic content and instrument tone colour. Temporal hearing: gap detection, time discrimination.	8	2

2	Mode of Hearing		15	
	2.1	Spatial hearing: sound localization, interaural time and level differences. Masking and critical bands in frequency perception. Auditory scene analysis: separating multiple sound sources.	7	2
	2.2	Emotional responses to sound and music. Brain regions involved in emotional sound processing. Cognitive load, multitasking, and auditory distractions. Auditory illusions: Shepard tones, McGurk effect, binaural beats.	8	3
3	Applications of Psychoacoustics		15	
	3.1	Audio compression: perceptual coding, lossy vs. lossless formats. Sound design for media: games, films, virtual and augmented reality.	7	4
	3.2	Use of psychoacoustics in noise control, architectural acoustics, and workplace design. Current trends and research in applied psychoacoustics.	8	4
4	4.1	Teacher Specific Content		

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Lectures, animations, diagrams, Audio demonstrations, sound simulations, interactive listening sessions • Simple auditory tests, demonstration of audiograms, discussions on hearing loss prevention. • Listening tests, analysis of music-induced emotions, illusion demonstrations.
Assessment Types	MODE OF ASSESSMENT Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) Theory Total Marks: 25 FA: Quiz, Assignments, Oral Presentations SA: Written Test
	B. End Semester Evaluation (ESE) Theory Total Marks: 50 , Duration: 1 hr MCQ Exam

MGU-B.VOC. (HONOURS)

References

1. Moore, Brian C. J. *An Introduction to the Psychology of Hearing*. 6th ed., Brill, 2012.
2. Zwicker, Eberhard, and Hugo Fastl. *Psychoacoustics: Facts and Models*. Springer, 1999.
3. Schnupp, Jan, Israel Nelken, and Andrew King. *Auditory Neuroscience: Making Sense of Sound*. MIT Press, 2011.
4. Alten, Stanley R. *Audio in Media*. London: Thomson Wadsworth, 2004.



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering				
Programme	MGU-B.Voc. (Honours) Sound Engineering				
Course Name	On the Job Training				
Type of Course	SDC				
Course Code	MG1SDCSEG103				
Course Level	100				
Course Summary	This course provides hands-on professional exposure to students through placement in real-world sound engineering environments such as recording studios, live sound setups, film post-production houses, broadcast stations, or media content units. It is designed to help students bridge academic knowledge with practical industry requirements, gain technical expertise, develop soft skills, and understand workplace dynamics, ethics, and standards. The training is supervised by industry mentors and monitored by faculty through periodic evaluations, reports, and presentations.				
Semester	1	Credits		2	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
	Practical sessions	0	0		
Pre- requisites, if any	Nil				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Demonstrate the ability to operate sound engineering equipment and software in professional work environments.	U	1,2,3,4,6,7, 8,9,10
2	Apply theoretical and technical knowledge in practical scenarios such	A	1,2,3,4,6,7, 8,9,10

	as recording, mixing, live sound, or post-production tasks.		
3	Exhibit professional conduct, communication, and teamwork during industry-based assignments.	I	1,2,3,4,5,6,7,8,9,10
4	Document and reflect on the learning process and tasks performed through reports, logs, and evaluations.	S	1,2,3,4,5,6,7,8,9,10

Assessment Types	MODE OF ASSESSMENT	
	Mode of Assessment	
	A. Continuous Comprehensive Assessment (CCA)	
	Total Marks: 15	
	Components	Marks
	Feedback from the hosting organization	5
	Internal Supervisor Feedback	10
	B. End Semester Evaluation (ESE)	
	Total Marks: 35	
	Components	Marks
	Presentation	10
	Report	10
	Viva Voice	15



SEMESTER 2

MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering			
Programme	MGU-B.Voc. (Honours) Sound Engineering			
Course Name	Analog and Digital Audio			
Type of Course	SDC			
Course Code	MG2SDCSEG100			
Course Level	100			
Course Summary	This course covers the basics of sound recording and reproduction, including both old and modern recording methods. It explains how magnetic tape machines work and how to maintain them. The course also introduces digital audio, covering topics like sampling, quantization, and conversion. Finally, it looks at different audio file formats, compression, digital connections, and synchronization, giving students a clear understanding of both analog and digital audio systems used in today's studios.			
Semester	2	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
	Lecture class, Hands-on Practicum sessions	4	0	0
Pre- requisites, if any	Nil			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Remember the basic principles and processes of sound recording and reproduction using phonographic, magnetic, and digital systems, including their advantages and disadvantages.	K	1,2,3,4,5, 6,7,8,9,10
2	Explain the working of magnetic recording systems, tape machines, tape heads, transport mechanisms, equalization, noise management, and modern tape emulation techniques.	U	1,2,3,4,5, 6,7,8,9,10
3	Understand digital electronics concepts including binary systems, logic gates, and digital audio principles such as sampling, Nyquist theorem, quantization, dithering, and analog-to-digital conversion.	U	1,2,3,4,5, 6,7,8,9,10
4	Identify and compare various audio file formats, compression methods, digital audio interconnection standards, synchronization systems, and digital audio level meters.	U	1,2,3,4,5, 6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	3	2	1	0	1	1	1	1	2
CO2	3	3	2	2	0	1	1	1	1	2
CO3	3	2	2	2	1	2	2	2	2	3
CO4	2	3	3	2	2	2	2	2	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Sound Recording and Reproduction		15	
	1.1	Phonographic, Flat Disk, Magnetic Tape	6	1
	1.2	Digital Audio Recording, Devices, Advantages and Disadvantages; Analog vs. Digital, Audio Recorders, Studio Recording, Limitations and Innovations	9	1
2	Magnetic recording and its media		15	
	2.1	The professional ATR, the magnetic tape head, the tape transport, equalization, monitoring modes	5	2
	2.2	Tape, Tape speed and Head configurations- print through, analogue tape noise	3	2
	2.3	Cleanliness, degaussing, C.L.A.S.P - functions and routing, tape emulation plugins.	7	2
3	Digital Electronics		15	
	3.1	The binary number system, Basic logic Gates and applications	5	3
	3.2	Digital audio- The theory, Sampling, Nyquist theorem, Aliasing, Bit depth, Quantisation, Quantization Error, Dither, Sample rate Conversion - Analog to digital, Digital to Analog, PCM	10	3
	Digital Audio File Formats, Interconnections, and Synchronization		15	
	4.1	Introduction to various audio file formats, Advantages and	7	4

4		disadvantages, features etc.- Compressed and Uncompressed audio, Lossy and lossless compression, AIFF and AIFF C format, BWF format, RIFF-WAVE format, FLAC, MP3.		
	4.2	Digital Audio Interconnection - SPDIF, AES/EBU, ADAT, TDIF, MADI, Network Audio	3	4
	4.3	Synchronization – Word Clock, Jitter, Digital Time-code, Interconnecting Digital Audio Recording Systems, Digital Audio Level Meters	5	4
5	5.1	Teacher Specific Content		

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Classroom Teaching: Easy-to-understand lessons with examples, diagrams, and videos to explain analog and digital audio. • Demonstrations: Live demos of tape machines, digital audio systems, audio formats, and recording setups. • Hands-on Practice: Students work with real equipment, record audio, connect systems, and practice using different formats and software. • Assessments: Regular quizzes, assignments, class discussions, and a final project or exam to test knowledge and skills.
Assessment Types	MODE OF ASSESSMENT Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) Theory Total Marks: 30 FA: Quiz, Oral Presentations, Assignments SA: Written Test

	B. End Semester Evaluation (ESE)	
	Theory	
	Total Marks: 70 , Duration: 2 hrs	
	Components	Marks
	Very short answer	10 out of 12 – 10 x 2 = 20 marks
	Short answer	6 out of 9 – 6 x 5 = 30 marks
	Essay	2 out of 3 – 10 x 2 = 20 marks

References

1. Alten, Stanley R. *Audio in Media*. London: Thomson Wadsworth, 2004.
2. Horowitz, Paul, and Winfield Hill. *The Art of Electronics*. 3rd ed., Cambridge University Press, 2015.
3. Ballou, Glen. *Handbook for Sound Engineers*. 5th ed., Routledge, 2015.
4. Watkinson, John. *The Art of Digital Audio*. 3rd ed., Focal Press, 2001.

MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering				
Programme	MGU-B.Voc. (Honours) Sound Engineering				
Course Name	Audio Signal Processing				
Type of Course	SDC				
Course Code	MG2SDCSEG101				
Course Level	100				
Course Summary	This course provides a comprehensive understanding of audio processing concepts, explores spectral, dynamics, and time-based processing tools and techniques, develops technical skills in applying various audio effects, and trains students in the practical use of DAW-based signal processing workflows.				
Semester	2	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
	Lecture class, Hands-on Practicum sessions	3	1	0	75
Pre- requisites, if any	Nil				

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Explain the fundamental principles of audio processing and signal routing	U	1,2,3,4,6,8,9,10
2	Apply various spectral and dynamics processing techniques for audio enhancement.	A	1,2,3,4,5,6,8,9,10

3	Use time-based effects creatively in music and post-production.	A	1,2,3,4,5,6,7,8,9,10
4	Operate DAW-based audio processing systems effectively for professional-level projects.	A	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	3	2	0	1	1	0	1	1	2
CO2	3	3	2	1	1	1	0	1	2	3
CO3	2	3	2	1	1	1	1	1	2	3
CO4	2	3	2	2	1	1	1	1	3	3

‘0’ is No Correlation, ‘1’ is Slight Correlation (Low level), ‘2’ is Moderate Correlation (Medium level) and ‘3’ is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Fundamentals of Audio Processing		15	
	1.1	Signal flow: Serial vs. Parallel Processing. Inserts, Sends, Y-cables, and Routing Techniques	5	1
	1.2	Bus Processing and Mid-Side Processing. Hardware vs. Software Processing	5	1
	1.3	Overview of: Dynamics Processing, Spectral Processing, Time-based Processing. Patchbays: Function and Types	5	1

2	Spectral & Harmonic Processing		15	
	2.1	Filters: High-Pass, Low-Pass, Band-Pass, Notch, Slope settings	4	2
	2.2	Equalizers: Bands: Low, Low-Mid, Mid, High-Mid, High. Types: Shelving, Semi-Parametric, Parametric, Graphic EQs in analog and digital mixers, EQ Tips for vocals, instruments Dynamic EQ functioning	6	2
	2.3	Harmonic Saturation: Harmonics: Odd & Even Tape, Tube Saturation, Analog Plugins Clipping processors, Distortion, Overdrive Guitar pedal effects	5	2
3	Dynamics & Time-Based Processing		15	
	3.1	Dynamics Processing: Compressors: Threshold, Ratio, Attack, Release, Gain, Side-chain, Knee Types: VCA, Opto, Vari-Mu, FET, Digital, Multi-band. Techniques: Parallel Compression, Ducking, De-Essing Expanders: Parameters & Usage, Limiters: Threshold, Ceiling, True-Peak, Noise Reduction: Noise Gate, De-noise plugins, Spectral Tools	8	3

	3.2	Time-Based Effects: Reverb: Types, Parameters, Routing, IR Reverbs, Reverb Plugins Delay: Types, Feedback, Sync to Tempo, Delay Time Calculation Chorus, Flanger, Phaser: Working principles and controls	7	3
		Practical Session	30	
4	4.1	Audio Processing in DAW: DAW Signal Chain: Insert vs Send Effect Bus Routing Master Bus Processing Parallel Processing Setup. Wet/Dry Control Management Plugin Chains for Mixing & Mastering Practical Tasks: Applying processing on multitrack sessions. Using real-world DAW plugins for dynamics, EQ, time-based effects. Processing different genres and scenarios (music, podcast, film audio, etc.) Submission of a fully processed project. Practical Evaluation: Hands-on tasks, individual processing projects, peer reviews.	30	4
5	5.1	Teacher Specific Content		

Teaching and Learning Approach	Project-Based DAW Training Assign real-world DAW projects that require setting up sessions, processing audio, mixing, and rendering final outputs to develop professional-level workflow skills.
Assessment Types	MODE OF ASSESSMENT Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) Practical Total Marks: 30 FA: Viva Voice SA: Competency tests in DAW
	B. End Semester Evaluation (ESE) Practical Total Marks: 70 , Duration: 3 hrs Lab Exam: 60 marks Viva: 10 marks

References

1. Huber, David Miles. *Modern Recording Techniques*. 9th ed. New York: Focal Press, 2017.
2. Owsinski, Bobby. *The Mixing Engineer's Handbook*. 5th ed. Milwaukee, WI: Hal Leonard Corporation, 2022.
3. Senior, Mike. *Mixing Secrets for the Small Studio*. 2nd ed. New York: Focal Press, 2017.
4. Izhaki, Roey. *Mixing Audio: Concepts, Practices, and Tools*. 3rd ed. New York: Routledge, 2018.



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering				
Programme	MGU-B.Voc. (Honours) Sound Engineering				
Course Name	DAW Fundamentals				
Type of Course	SDC				
Course Code	MG2SDCSEG102				
Course Level	100				
Course Summary	This course introduces the basics and advanced features of Pro Tools, helps students develop skills in recording, editing, mixing, and exporting audio, applies DAW techniques to real-world music and post-production projects, and builds hands-on proficiency for professional audio production.				
Semester	2	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
	Lecture class, Hands-on Practicum sessions	3	1	0	75
Pre- requisites, if any	Nil				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the interface, tools, and workflow of DAW	U	1,2,3,4,5, 6,7,8,9,10
2	Apply recording, editing, and processing techniques using DAW.	A	1,2,3,4,5, 6,7,8,9,10

3	Demonstrate knowledge of advanced mixing, automation, and export features	A	1,2,3,4,5,6,7,8,9,10
4	Execute complete audio production projects using Pro Tools with practical proficiency.	A	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	2	2	2	2	1	1	1	1	1	2
CO2	2	3	2	2	2	1	1	1	2	3
CO3	3	3	2	2	2	1	1	1	2	3
CO4	3	3	2	2	3	1	1	1	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Introduction to DAW		15	
	1.1	Overview of DAWs and their role in audio production. Introduction to Pro Tools software versions and system requirements.	3	1
	1.2	Pro Tools interface: Edit Window, Mix Window, Transport Bar, Menus. Session setup: Creating and managing sessions.	7	1
	1.3	File types, session templates, and storage options. Audio formats and bit depth, sample rate consideration	5	1

	Recording and Editing in Pro Tools		15	
	2.1	Setting up audio interfaces and input/output routing. Track types: Audio, Instrument, MIDI, Aux, Master.	5	2
	2.2	Signal flow and gain staging. Audio and MIDI recording techniques.	3	2
	2.3	Editing tools: Trim, Selector, Grabber, Smart Tool. Editing techniques: Cutting, Fading, Crossfades, Time Stretch, Elastic Audio. Loop recording and playlist management.	7	2
	Mixing, Processing & Export		15	
	3.1	Introduction to mixing workflow in Pro Tools. Using Inserts, Plugins, and Virtual Instruments.	5	3
3	3.2	Equalization, Compression, Reverb, Delay, Modulation. Aux sends and returns, sub-mixing, buses. Automation: Volume, Pan, Plugin parameters. Mastering basics: Metering, loudness, normalization. Exporting and bouncing sessions for various formats.	10	3

4	Practical Session		30	
	4.1	Hands-on exercises: Session creation, recording, and editing. Track editing, comping, and arranging.	20	4
	4.2	Full mixdown exercises with effects, routing, and automation.	10	4
5	5.1	Teacher Specific Content		

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) - Lectures, software demonstrations, visual guides - Step-by-step software practice, analysis of professional mix sessions - Students will submit fully mixed and edited projects demonstrating their skills.
Assessment Types	MODE OF ASSESSMENT Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) Practical Total Marks: 30 FA: Viva Voice SA: Competency tests in DAW
	B. End Semester Evaluation (ESE) Practical Total Marks: 70 , Duration: 3 hrs Lab Exam: 60 marks Viva Voice :10 marks

References

1. Cook, David. *Pro Tools 101: An Introduction to Pro Tools*. Avid Learning Series, Latest Edition.
2. Gibson, Bill. *Pro Tools for Music Production: Recording, Editing and Mixing*. Focal Press, 2005.
3. Izhaki, Roey. *Mixing Audio: Concepts, Practices and Tools*. Focal Press, 2018.
4. Avid Technology. *Pro Tools Reference Guide*. (Latest Online Documentation)



MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering				
Programme	MGU-B.Voc. (Honours) Sound Engineering				
Course Name	Music Composition				
Type of Course	MDC				
Course Code	MG2MDCSEG100				
Course Level	100				
Course Summary	This course introduces the fundamentals of music composition, focusing on melody, harmony, rhythm, and structure. Students will learn to create original pieces using traditional notation and digital tools, while exploring various styles and techniques to develop their own compositional voice.				
Semester	2	Credits		3	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
	Lecture class, Hands-on Practicum sessions	2	1	0	60
Pre- requisites, if any	Nil				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the sequence of composing original songs by developing ideas from inspiration through structured songwriting techniques.	U	1,2,3,4,5, 6,7,8,9,10

2	Make soundscapes that communicate emotion, tell stories, and connect with listeners. Write compelling lyrics	A	1,2,3,4,5, 6,7,8,9,10
3	Make effective melodies, harmonies, and arrangements that reflect genre, mood, and artistic intent.	A	1,2,3,4,5, 6,7,8,9,10
4	Present and refine songs through demo production, performance, and constructive feedback for artistic growth.	A	1,2,3,4,5, 6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	2	2	2	1	1	1	1	2	1	2
CO2	2	1	2	3	1	2	2	3	1	2
CO3	2	2	3	2	1	1	1	2	1	2
CO4	2	2	2	3	2	1	1	2	2	3

‘0’ is No Correlation, ‘1’ is Slight Correlation (Low level), ‘2’ is Moderate Correlation (Medium level) and ‘3’ is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Song Structure		15	
	1.1	Learn core song structures and their functions (verse, chorus, bridge, hook).	3	1
	1.2	Discover sources of inspiration and develop compelling song ideas.	7	1
	1.3	Cultivate creative flow and overcome writer's block	5	1
2	Lyrics		15	
	2.1	Write authentic, emotionally engaging lyrics using storytelling and imagery.	5	2
	2.2	Apply rhyme schemes, meter, and lyrical flow for musicality.	3	2

	2.3	Edit and refine lyrics for clarity, impact, and emotional resonance	7	3
3	Practical		30	
	3.1	Compose expressive melodies and harmonies using scales, intervals, and emotional tone.	15	3
	3.2	Shape rhythm, groove, and arrangement by exploring genre styles, dynamics, and collaboration. Bring songs to life through demo recording, performance, feedback, and personal artistic growth.	15	4
4	4.1	Teacher Specific Content		

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Theory Classes: Easy-to-understand lectures • Demonstrations: Live demonstrations of programming • Assessments by conducting contests
Assessment Types	MODE OF ASSESSMENT Mode of Assessment A. Continuous Comprehensive Assessment (CCA) Practical Total Marks: 25 FA: Viva Voice SA: Practical test B. End Semester Evaluation (ESE) Practical Total Marks: 50 , Duration: 3 hrs Lab Exam: 40 marks Viva: 10 marks

References

1. Schoenberg, Arnold. *Fundamentals of Musical Composition*. London: Faber & Faber, 1967.
2. Stevenson, David, and Jeffrey Ainis. *Composing Music: A New Approach*. Milwaukee, WI: Hal Leonard Corporation, 1991.
3. Adler, Samuel. *The Study of Orchestration*. 4th ed. New York: W. W. Norton & Company, 2016.



MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering			
Programme	MGU-B.Voc. (Honours) Sound Engineering			
Course Name	On the Job Training			
Type of Course	SDC			
Course Code	MG2SDCSEG103			
Course Level	100			
Course Summary	This course provides hands-on professional exposure to students through placement in real-world sound engineering environments such as recording studios, live sound setups, film post-production houses, broadcast stations, or media content units. It is designed to help students bridge academic knowledge with practical industry requirements, gain technical expertise, develop soft skills, and understand workplace dynamics, ethics, and standards. The training is supervised by industry mentors and monitored by faculty through periodic evaluations, reports, and presentations.			
Semester	2	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
	Practical sessions	0	0	60
Pre-requisites, if any	Nil			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Demonstrate the ability to operate sound engineering equipment and software in professional work environments.	U	1,2,3,4,6,7, 8,9,10
2	Apply theoretical and technical knowledge in practical scenarios such	A	1,2,3,4,6,7, 8,9,10

	as recording, mixing, live sound, or post-production tasks.		
3	Exhibit professional conduct, communication, and teamwork during industry-based assignments.	I	1,2,3,4,5,6,7,8,9,10
4	Document and reflect on the learning process and tasks performed through reports, logs, and evaluations.	S	1,2,3,4,5,6,7,8,9,10

Assessment Types	MODE OF ASSESSMENT	
	Mode of Assessment	
	A. Continuous Comprehensive Assessment (CCA)	
	Total Marks: 15	
	Components	Marks
	Feedback from the hosting organization	5
	Internal Supervisor Feedback	10
	B. End Semester Evaluation (ESE)	
	Total Marks: 35	
	Components	Marks
	Presentation	10
	Report	10
	Viva Voice	15

SEMESTER 3



MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering			
Programme	MGU-B.Voc. (Honours) Sound Engineering			
Course Name	Music Architecture			
Type of Course	SDC			
Course Code	MG3SDCSEG200			
Course Level	200			
Course Summary	This course provides a foundational understanding of music theory, covering essential concepts such as musical notation, rhythm, scales, intervals, chords, and harmony. Students will learn to read and write music using the staff, clefs, and time signatures, and explore the structure of major and minor scales, key signatures, and chord progressions. The course also introduces basic elements of musical form and arrangement, including the SATB (Soprano, Alto, Tenor, Bass) structure. Through practical exercises and listening activities, students will develop the skills needed to analyse and create music confidently.			
Semester	3	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
	Lecture class, Hands-on Practicum sessions	4	0	0
Pre- requisites, if any	Nil			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Demonstrate an understanding of the history, evolution, and stylistic characteristics of Western classical music and popular music genres through active listening and analysis.	U	1,2,3,4,6,7,8,9,10
2	Identify and understand orchestral instruments and their sections, explaining the structure and setup of classical orchestras.	U	1,2,3,4,5,6,7,8,9,10
3	Understand basic music notation, including clefs, note values, rhythmic patterns, and musical symbols.	U	1,2,3,4,5,6,7,8,9,10
4	Apply foundational music theory concepts such as scales, chords, intervals, key signatures, and harmony to analyze and construct basic musical arrangements.	A	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	3	2	1	0	1	1	1	1	2
CO2	2	3	2	1	1	1	1	1	1	2
CO3	2	3	2	2	1	1	1	1	2	2
CO4	2	3	2	1	2	1	1	1	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Music history		15	
	1.1	Foundations of Music: Introduces the concept of music and provides a brief overview of the history of Western classical music, covering major eras including Baroque,	5	1

		Renaissance, Classical, Romantic, and Modern.		
	1.2	Classical Music Elements: Explores key classical music styles such as duet, quartet, sonata, symphony, and opera, along with notable classical composers who shaped each era.	5	1
	1.3	Listening and Popular Genres: Includes listening sessions to help students recognize classical music pieces and introduces popular contemporary genres like rock, pop, blues, jazz, hip-hop, and progressive music.	5	1
	Musical Instruments		15	
	2.1	Overview of the Classical Orchestra: Introduces the structure, evolution, and types of classical orchestras, including symphony and chamber orchestras.	3	2
2	2.2	Orchestral Setup: Explains the standard layout and arrangement of instruments within an orchestra for optimal sound balance and performance.	7	2
	2.3	Instrument Families: Categorizes orchestral instruments into four main sections—Strings (violin, viola, cello, double bass), Winds (flute, oboe), Brass (trumpet, trombone, tuba, horns), and Percussion—highlighting their roles and characteristics.	5	3
	Music Notation		15	
4	3.1	Music Notation and Note Values: Covers the basics of reading music using the staff, treble and bass clefs, note names, middle C, half/whole steps, accidentals, and various note values (e.g., semibreve to demi-semiquaver), including	5	3

		rests, dotted notes, triplets, bar-lines, and other musical symbols.		
	3.2	Pitch, Instruments, and Rhythm: Teaches how to map musical notes to piano keys, understand the pitch ranges of different instruments, and recognize rhythmic elements like pulse, meter, and time signatures	6	4
	3.3	Scales, Chords, and Harmony: Introduces major, minor, and chromatic scales, key signatures, intervals, modes, triads, chord inversions, and progressions—laying the foundation for understanding harmony and basic musical arrangement.	4	4

		Indian music	15	
	4.1	Introduce the concepts of <i>raga</i> (melodic framework) and <i>tala</i> (rhythmic cycle), along with basic swaras (notes) and their emotional and spiritual significance.	3	1
4	4.2	Explore the two main systems— <i>Hindustani</i> (North Indian) and <i>Carnatic</i> (South Indian) music—highlighting their key differences, vocal and instrumental forms, and notable composers	7	1
	4.3	Discuss the role of Indian music in religious, social, and cultural practices, including its evolution, connection to dance and theatre, and influence on contemporary music.	5	2
5	5.1	Teacher Specific Content		

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) 1. Theory Classes: Easy-to-understand lectures 2. Using technology for familiarization of musical instruments
Assessment Types	MODE OF ASSESSMENT Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) Practical Total Marks: 30 FA: Assignments, Oral Presentations, Viva Voice SA: Practical test
	B. End Semester Evaluation (ESE) Practical Total Marks: 70 , Duration: 3 hrs Lab Exam: 60 marks Viva Voice: 10 marks

References

1. Kostka, Stefan, and Dorothy Payne. *Tonal Harmony*. 8th ed. New York: McGraw-Hill Education, 2017.
2. Laitz, Steven G. *The Complete Musician: An Integrated Approach to Tonal Theory, Analysis, and Listening*. 4th ed. New York: Oxford University Press, 2016.
3. Pilhofer, Michael, and Holly Day. *Music Theory for Dummies*. 4th ed. Hoboken, NJ: Wiley, 2021.



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering			
Programme	MGU-B.Voc. (Honours) Sound Engineering			
Course Name	Sampling and Synthesis			
Type of Course	SDC			
Course Code	MG3SDCSEG201			
Course Level	200			
Course Summary	This course introduces the fundamentals of audio sampling and sound synthesis. Students learn how to capture, edit, and manipulate sound samples, and explore various synthesis techniques including subtractive, additive, and FM synthesis. Practical sessions focus on using digital audio workstations and synthesizers to create original sounds and music compositions.			
Semester	3	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
	Lecture class, Hands-on Practicum sessions	3	1	0
Pre- requisites, if any	Nil			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand a synthesizer is, distinguish between analog and digital synthesis, and describe the historical development and applications of various synthesis techniques in music.	U	1,2,3,4,8,10

2	Demonstrate knowledge of waveform types, signal flow, and key synthesizer components (VCO, VCF, VCA, EG, LFO), and apply synthesis methods such as subtractive, additive, FM, and wavetable synthesis to create custom sounds.	A	1,2,3,4,5, 8,9,10
3	Apply key sampling concepts such as sample rate, bit depth, looping, pitch shifting, and multi-sampling to build original sample-based instruments and libraries.	A	1,2,3,4,5, 8,9,10
4	Create original synthesizer patches (e.g., leads, pads, bells) using VST instruments and develop drum or instrument sample libraries using multi-layered sampling techniques and digital audio editing.	C	1,2,3,4,5, 6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	2	1	1	0	0	0	1	0	2
CO2	3	3	2	1	1	0	0	1	0	3
CO3	2	3	2	1	1	0	0	1	1	3
CO4	2	2	2	3	2	1	1	2	2	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT **MGU-B.VOC. (HONOURS)**

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Synthesizer basics		15	
	1.1	Understand what a synthesizer is and explore the evolution of synthesizers from analog to digital.	5	1
	1.2	Learn about different synthesis methods including subtractive, additive, wavetable, FM, sample replay, and physical modelling.	5	1

	1.3	Explore software synthesis and the various uses of synthesizers in music production and sound design.	5	2
2	Analog vs Digital		15	
	2.1	Fundamentals of Analog and Digital Synthesizers Explore analog synthesizers, Fourier analysis, basic waveforms (sine, square, triangle, sawtooth), and core sound properties like amplitude, pitch, timbre, loudness, and envelope shaping.	6	2
	2.2	Synth Types and Components: Learn about different synthesizer types (modular, analog, digital, sample-playback, software), monophonic vs. polyphonic synths, and key components such as oscillators (VCO), filters (VCF), envelope generators, amplifiers (VCA), LFOs, and wave shapers.	4	2
	2.3	Synthesis Methods and Tools: Understand synthesis techniques including subtractive, additive, AM, FM, ring modulation, physical modelling, granular, wavetable, and drum synthesis, along with popular hardware models and software plug-ins.	5	2
3	Sampling		15	
	3.1	Understand the difference between synthesizers and samplers, and learn the fundamentals of sampling including sample rate and bit depth.	5	3

	3.2	Explore key methods such as looping, envelope generation, one-shot samples, pitch shifting, and multi-sampling. Learn how to edit and process samples for drum sampling, create sample libraries, and apply them in music production	10	3
4	Practical Session		30	
	4.1	Synth Patch Creation: Use a synth VSTi to explore basic routing, waveform selection, AMP and LFO sections, and design custom patches like leads, pads, and bells	8	4
	4.2	Drum Sampling Techniques: Record and edit one-shot drum samples with velocity layers, assign samples to samplers, and fine-tune for performance.	7	4
	4.3	Instrument Sampling and Library Building: Record instrument performances, apply key-mapping, velocity layers, envelope shaping, pitch shifting, and looping to create versatile sample libraries.	15	4
5	5.1	Teacher Specific Content		

Syllabus

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Theory Classes: Easy-to-understand lectures with diagrams, videos, and real-life examples to explain basic electronics, digital audio, and sound recording concepts. • Practical Sessions: Hands-on practice with using audio equipment.
Assessment Types	MODE OF ASSESSMENT Mode of Assessment

	A. Continuous Comprehensive Assessment (CCA) Practical Total Marks: 30 FA: Viva Voice, Assignments SA: Practical test
	B. End Semester Evaluation (ESE) Practical Total Marks: 70, Duration: 3 hrs Project Evaluation: 60 marks Viva: 10 marks

References

1. Strange, Allen. *Electronic Music: Systems, Techniques, and Controls*. 2nd ed. Dubuque, IA: Kendall Hunt Publishing, 1983.
2. Welsh, Fred. *Welsh's Synthesizer Cookbook: Synthesizer Programming, Sound Analysis, and Universal Patch Book*. 2nd ed. Port St. Lucie, FL: Bradshaw Publishing, 2006.
3. Watkinson, John. *The Art of Digital Audio*. 3rd ed. Oxford: Focal Press, 2001.



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering				
Programme	MGU-B.Voc. (Honours) Sound Engineering				
Course Name	Live Sound Fundamentals				
Type of Course	SDC				
Course Code	MG3SDCSEG202				
Course Level	200				
Course Summary	This course introduces the fundamentals of live sound equipment, system setup, and operation, develops understanding of signal flow, audio connections, and mixer functions, explores speaker systems, amplification, and PA tuning for different venues, and provides hands-on experience with digital mixing consoles and live sound operations.				
Semester	3	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
	Lecture class, Hands-on Practicum sessions	3	1	0	75
Pre- requisites, if any	Nil				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the basics of sound, dB scales, equipment, and safety in live sound setups.	U	1,2,3,4,5,6,7,8,9,10
2	Apply proper connection techniques, signal flow management, and live sound miking techniques	A	1,2,3,4,5,6,7,8,9,10
3	Explain the working of speakers, amplifiers, PA systems, and implement system calibration.	U	1,2,3,4,5,6,7,8,9,10
4	Operate digital mixing consoles for live events, virtual sound checks, and multi-track recording.	A	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	3	2	1	1	2	1	1	1	2
CO2	3	3	3	2	2	2	1	1	2	3
CO3	2	3	2	1	1	2	2	1	1	2
CO4	3	3	3	2	2	1	1	1	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

MGU-B.VOC. (HONOURS)

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Fundamentals of Live Sound & Equipment		15	
	1.1	Sound Basics: dB scales, Inverse-Square Law, SNR, Dynamic Range, Headroom	4	1
	1.2	Live Sound Equipment Overview: Microphones: Dynamic, Condenser, Applications. Mixers: Analog vs Digital, Console Types. Outboard Gear: EQ,	7	1

		Compressors, Effects Processors. Speakers: Types and Applications. Cables and Connectors. Basic Electrical Safety.		
	1.3	Preparing a Tech Rider for Live Shows	4	1
2	Audio Connections, Signal Flow & Miking		15	
	2.1	Power and Audio Connections. Mono vs Stereo, Series & Parallel, Impedance Matching, Balanced/Unbalanced Connections	3	2
	2.2	DI Box usage. Analog Mixer Operation: Signal Flow, Channel Strip Functions Gain Staging, Filters, EQ Types. AUX Sends, PAN, Faders, Metering (VU, Peak) Inserts, Sends, External Gear Integration Grouping: VCA, Group Master, Aux Master	5	2
	2.3	Miking Techniques: Drums, Vocal, Instrument Miking Sound Check Procedures	7	2
3	Speakers, Amplifiers & PA Tuning		15	
	3.1	Speaker Systems: Active vs Passive, Horn & Cone Types Subwoofers, Line Arrays, Monitor Setup, 2/3 Way Systems. Crossovers & Crossover Points	5	3

	3.2	Amplifiers: Preamp & Power Amp Basics. Amplifier Classes & Specifications. Cable Types: Microphone, Instrument, Speaker, Y-cable, Specialized Cables	4	3
	3.3	PA System Tuning: Room Calibration, Calibration Tools, Software Solutions Speaker Delays and Time Alignment	6	3
	Practical Session		30	
4	4.1	Digital Console Introduction: Advantages and Features. I/O Connections	10	4
	4.2	Console Operations: Patching, Signal Flow Management. EQ, Dynamics, FX Application. Grouping, DCA Use. Saving & Loading pre-sets	10	4
	4.3	Interfacing with Computer Systems: Plugin Integration. Multi-track Recording via Digital Mixer. Virtual Sound Check Procedures	10	4
5	5.1	Teacher Specific Content		

Teaching and Learning Approach	Classroom Procedure (Mode of transaction)
	Interactive Lectures & Visual Aids Core concepts such as live sound equipment, signal flow, speaker systems, and amplification are introduced through lectures supported by diagrams, videos, and system walkthroughs. Hands-On Equipment Demonstration Students get direct exposure to mixers, microphones, cables, speakers, and amplifiers with guided demonstrations of setup, routing, and operation.

	3. Lab-Based Practice Sessions Practical sessions allow students to build and operate live sound setups, explore signal paths, perform sound checks, and calibrate PA systems in simulated or real environments. 4. Project-Based Learning & Group Activities Students work in teams to plan and execute live sound setups using digital consoles, manage roles during mock events, and troubleshoot real-time technical challenges.
Assessment Types	MODE OF ASSESSMENT Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) Practical Total Marks: 30 FA: Viva Voice, Assignment SA: Competency tests in Audio mixer
	B. End Semester Evaluation (ESE) Practical Total Marks: 70, Duration: 3 hrs Lab Exam: 60 marks Viva: 10 marks

Syllabus

References

1. Davis, Don, and Carolyn Davis. *Sound System Engineering*. 4th ed. New York: Focal Press, 2013.
2. Woram, John. *The Recording Studio Handbook*. Indianapolis, IN: Howard W. Sams & Co., 1989.
3. Huber, David Miles. *Modern Recording Techniques*. 9th ed. New York: Focal Press, 2017.
4. Eargle, John. *Handbook of Recording Engineering*. 4th ed. New York: Springer, 2006.



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering			
Programme	MGU-B.Voc. (Honours) Sound Engineering			
Course Name	Sound, Society & Technology			
Type of Course	MDC			
Course Code	MG3MDCSEG200			
Course Level	200			
Course Summary	This course introduces the basic concepts of sound from both scientific and traditional perspectives, explores the role of sound in Kerala's ecology, culture, and heritage, creates awareness about its importance in health, society, and sustainable development, and encourages responsible and ethical sound practices using both modern and traditional knowledge systems.			
Semester	3	Credits		Total Hours
Course Details	Learning Approach Lecture class , demo sessions	Lecture 3	Practical 0	
			OJT 0	45
Pre- requisites, if any	Nil			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the scientific and traditional foundations of sound and hearing	U	1,2,3,4,5, 6,7,8,9,10
2	Analyze the role of sound in Kerala's ecology, architecture, culture, and performing arts.	An	1,2,3,4,5, 6,7,8,9,10

3	Apply knowledge of sound technology and modern media in Kerala's social and cultural context.	A	1,2,3,4,5,6,7,8,9,10
4	Understand the impact of sound on health, environment, and society, and explore sustainable sound practices.	U	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	3	3	2	1	2	2	2	1	3
CO2	3	2	3	3	2	3	3	2	2	3
CO3	2	2	3	3	2	3	3	2	3	3
CO4	3	3	2	2	1	3	3	3	2	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Sound in Kerala's Culture and Ecology		15	
	1.1	Traditional Kerala Knowledge: Temple sound practices: Chenda Melam, Sopana Sangeetham. Healing with sound in Ayurveda: Use of sound, silence, and natural soundscapes. Panchabhoota theory: Relationship between sound and five elements	3	1
	1.2	Kerala's Soundscapes: Monsoon rains, Western Ghats bird calls, temple bells, boat race drumming.	7	2

		Acoustic Architecture: Nalukettu houses, temples, churches, mosques — sound reflection and resonance.		
	1.3	Performing Arts: Kathakali, Theyyam, Koodiyattam, Mappila Paattu, Margamkali and so on— structure, rhythm, instruments. Oral Traditions: Folk narratives, storytelling, and preservation of cultural memory through sound.	5	2
	Modern Sound Technologies and Applications		15	
	2.1	Technological Basics: Microphones, loudspeakers, recording devices. Digital audio: Sampling, bit depth, file formats, compression.	5	3
	2.2	Sound in Media: Radio broadcasting (Akashvani Kerala), cinema sound, podcasts, social media. Community radio and rural knowledge sharing.	3	3
	2.3	Ethical Concerns: Noise pollution, hearing conservation, responsible listening practices.	7	3
	Sound, Sustainability and Society		15	
3	3.1	Environmental Impact: Noise pollution in urban Kerala and pilgrimage sites. Eco-acoustics of Kerala forests (Silent Valley, Periyar).	5	4

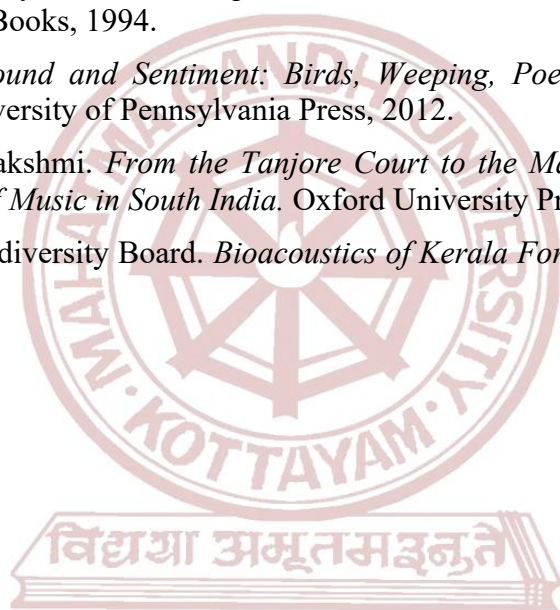
	3.2	Health and Wellness: Sound in Ayurveda, meditation, and modern sound therapy.	5	4
	3.3	Sustainable Cultural Preservation: Documenting disappearing sound traditions. Eco-friendly practices in festivals and sound events. Use of AI & machine learning in oral tradition preservation.	5	4
4	4.1	Teacher Specific Content		

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) 1. Lectures, audio-visual demonstrations, case studies on Kerala rituals, short field recordings 2. Live demonstrations, documentary screenings, analysis of traditional performances, group discussions 3. Studio visits, hands-on exposure to recording equipment, guest lectures from media professionals, student presentations.
Assessment Types	MODE OF ASSESSMENT Mode of Assessment A. Continuous Comprehensive Assessment (CCA) Theory Total Marks: 25 FA: Quiz, Assignment, Oral Presentations SA: Written Test

	B. End Semester Evaluation (ESE) Theory Total Marks: 50, Duration: 1 hr MCQ Exam
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References

1. Schafer, R. Murray. *The Soundscape: Our Sonic Environment and the Tuning of the World*. Destiny Books, 1994.
2. Feld, Steven. *Sound and Sentiment: Birds, Weeping, Poetics, and Song in Kaluli Expression*. University of Pennsylvania Press, 2012.
3. Subramanian, Lakshmi. *From the Tanjore Court to the Madras Music Academy: A Social History of Music in South India*. Oxford University Press, 2006.
4. Kerala State Biodiversity Board. *Bioacoustics of Kerala Forests* (Reports & Research Papers).



MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering			
Programme	MGU-B.Voc. (Honours) Sound Engineering			
Course Name	On the Job Training			
Type of Course	SDC			
Course Code	MG3SDCSEG203			
Course Level	200			
Course Summary	This course provides hands-on professional exposure to students through placement in real-world sound engineering environments such as recording studios, live sound setups, film post-production houses, broadcast stations, or media content units. It is designed to help students bridge academic knowledge with practical industry requirements, gain technical expertise, develop soft skills, and understand workplace dynamics, ethics, and standards. The training is supervised by industry mentors and monitored by faculty through periodic evaluations, reports, and presentations.			
Semester	3	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
	Practical sessions	0	0	60
Pre-requisites, if any	Nil			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Demonstrate the ability to operate sound engineering equipment and software in professional work environments.	U	1,2,3,4,6,7, 8,9,10
2	Apply theoretical and technical knowledge in practical scenarios such	A	1,2,3,4,6,7, 8,9,10

	as recording, mixing, live sound, or post-production tasks.		
3	Exhibit professional conduct, communication, and teamwork during industry-based assignments.	I	1,2,3,4,5,6,7,8,9,10
4	Document and reflect on the learning process and tasks performed through reports, logs, and evaluations.	S	1,2,3,4,5,6,7,8,9,10

Assessment Types	MODE OF ASSESSMENT	
	Mode of Assessment	
	A. Continuous Comprehensive Assessment (CCA)	
	Total Marks: 15	
	Components	Marks
	Feedback from the hosting organization	5
	Internal Supervisor Feedback	10
	B. End Semester Evaluation (ESE)	
	Total Marks: 35	
	Components	Marks
	Presentation	10
	Report	10
	Viva Voice	15

SEMESTER 4



MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering				
Programme	MGU-B.Voc. (Honours) Sound Engineering				
Course Name	Media Management				
Type of Course	SDC				
Course Code	MG4SDCSEG200				
Course Level	200				
Course Summary	This course helps students understand how different media organizations work. They will learn how to apply basic management skills in media production and operations. The course also covers how media companies make money, build brands, and use marketing. Students will explore how digital technology is changing the media industry and how to adapt to new trends.				
Semester	4	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
	Lecture Class , demo sessions	4	0	0	
Pre- requisites, if any	Nil				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the structure and functions of various media organizations.	U	1,2,3,4,6, 7,8,9,10
2	Apply management principles in media production and operations.	A	1,2,3,4,6, 7,8,9,10

3	Understand revenue models, branding, and marketing strategies in media businesses.	U	1,2,3,4,6,7,8,9,10
4	Understand the impact of digital transformation and emerging technologies on media.	U	1,2,3,4,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	2	2	2	1	2	1	1	1	2
CO 2	2	2	3	2	3	2	1	2	2	2
CO 3	2	2	3	2	3	2	2	2	2	2
CO 4	3	3	3	3	2	2	2	2	2	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No
1	Foundations of Media Management		15	
	1.1	Introduction to Media Industry (Print, Broadcast, Digital, New Media), Nature and Structure of Media Organizations	6	1
	1.2	Public vs. Private Media Ownership, Basic Principles of Management (Planning, Organizing, Staffing, Directing, Controlling)	7	1
	1.3	Leadership and Decision-Making in Media	2	1
2	Content & Sound Management		15	
	2.1	Programming and Content Strategy (Radio, TV, OTT), Sound Management in Media: Studios, Equipment, Quality Control	6	2

	2.2	Production Copyright, Licensing, and Intellectual Property in Media	6	2
	2.3	Case Studies: Radio Station or Podcast Management	3	2
	Media Business and Marketing		15	
3	3.1	Revenue Models: Advertising, Subscriptions, Sponsorships, Pay-per-view,	7	3
	3.2	Sales and Distribution of Media Products, Branding and Audience Targeting	6	3
	3.3	Media Marketing: Traditional vs. Digital Campaigns	2	3
	Digital Media & Emerging Trends		15	
4	4.1	Convergence of Media Platforms, Social Media Management Tools & Strategy	3	4
	4.2	AI and Automation in Media Editing and Content Personalization	5	4
	4.3	Ethics and Legal Issues in the Digital Age	7	4
5	5.1	Teacher Specific Content		

Syllabus

Teaching and Learning Approach	Teaching Methods (Theory) • Lectures on media structures, management, and business models • Case studies of real media companies (e.g., TV channels, digital platforms, newspapers) • Guest lecturers from media professionals • Discussions on current trends and challenges in media industries
Assessment Types	MODE OF ASSESSMENT Mode of Assessment

	A. Continuous Comprehensive Assessment (CCA) Theory Total Marks: 30 FA: Quiz, Oral Presentations SA: Written Test							
	B. End Semester Evaluation (ESE) Theory Total Marks: 70 , Duration: 2 hrs							
	<table><tr><th>Components</th><th>Marks</th></tr><tr><td>Very short answer</td><td>10 out of 12 – 10 x 2 = 20 marks</td></tr><tr><td>Short answer</td><td>6 out of 9 – 6 x 5 = 30 marks</td></tr><tr><td>Essay</td><td>2 out of 3 – 10 x 2 = 20 marks</td></tr></table>	Components	Marks	Very short answer	10 out of 12 – 10 x 2 = 20 marks	Short answer	6 out of 9 – 6 x 5 = 30 marks	Essay
Components	Marks							
Very short answer	10 out of 12 – 10 x 2 = 20 marks							
Short answer	6 out of 9 – 6 x 5 = 30 marks							
Essay	2 out of 3 – 10 x 2 = 20 marks							

References

1. Picard, Robert G. *The Economics and Financing of Media Companies*. 2nd ed. New York: Fordham University Press, 2011.
2. McQuail, Denis, and Mark Deuze. *McQuail's Media and Mass Communication Theory*. 7th ed. London: Sage Publications, 2020.
3. Eastman, Susan Tyler, Douglas A. Ferguson, and Robert A. Klein. *Media Programming: Strategies and Practices*. 9th ed. Boston: Cengage Learning, 2012.
4. Bielby, Denise D., and C. Lee Harrington. *Global TV: Exporting Television and Culture in the World Market*. New York: NYU Press, 2008.



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering			
Programme	MGU-B.Voc. (Honours) Sound Engineering			
Course Name	MIDI and Music Programming			
Type of Course	SDC			
Course Code	MG4SDCSEG201			
Course Level	200			
Course Summary	This course introduces students to the fundamentals of music production using Digital Audio Workstations (DAWs) and MIDI (Musical Instrument Digital Interface) technology. Students will learn how to sequence, edit, and arrange music using virtual instruments and audio recordings. Key topics include MIDI programming, signal flow, audio effects, mixing techniques, and basic sound design. The course emphasizes hands-on experience with industry-standard software, enabling students to create professional-quality music tracks. By the end of the course, students will be able to produce, edit, and export original music projects using both MIDI and audio workflows.			
Semester	4	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
	Lecture class, Hands-on Practicum sessions	3	1	0
Pre- requisites, if any	Nil			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Demonstrate an understanding of MIDI technology, including its structure, communication protocol, hardware interfaces, and system configurations such as daisy-chaining and multiport networks.	U	1,2,3,4,8,10
2	Apply MIDI knowledge to operate electronic instruments, use MIDI controllers, and perform basic sequencing tasks including recording, tempo setting, and managing MIDI messages and channels.	A	1,2,3,4,5,8,9,10
3	Set up a digital audio workstation (DAW), connect MIDI devices, and utilize virtual instruments, sample libraries, and plugin formats (VST/AU/AAX) for music programming.	A	1,2,3,4,5,8,9,10
4	Apply music pre-production techniques including song structuring, setting tempo and pitch, arranging chords, and configuring MIDI sessions effectively.	A	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	2	1	1	0	0	0	1	0	2
CO2	3	3	2	1	1	0	0	1	0	3
CO3	2	3	2	1	1	0	0	1	1	3
CO4	2	2	2	3	2	1	1	2	2	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Synthesizer Basics		15	
	1.1	Understanding MIDI and Its Role: Learn what MIDI is, how it functions in music production environments, and how it interfaces with computers and external devices.	5	1
	1.2	MIDI Connectivity and Setup: Explore MIDI system connections, including cables, jacks, phantom power, wireless options, and common configurations like daisy chains and multiport networks.	5	1
	1.3	MIDI Protocol and Communication: Understand the MIDI specification—covering messages, channels, modes, and the distinction between channel voice and system messages.	5	2
2	Programming fundamentals		15	
	2.1	Music Programming Basics: Introduction to DAWs, connecting audio/MIDI devices, and working with virtual instruments, loops, sample libraries, and sound banks.	3	2
	2.2	Pre-Production Essentials: Learn how to structure a song, decide tempo, time signature, and pitch, and write basic arrangements including chords and score.	7	2
	2.3	Session Setup and MIDI Integration: Set up programming sessions by configuring MIDI devices, metronome settings, and understanding plugin formats (VST/AU/AAX) and multichannel virtual instruments.	5	3

3	Sampling		15	
	3.1	MIDI Recording Tools Understand how to use the piano roll and drum roll interfaces to record and edit MIDI notes, including control over note length and layout.	7	3
	3.2	Advanced MIDI Recording Techniques: Learn methods like MIDI overdub vs merge, punch in/out, step recording, groove programming, looping, and handling track delays and latency issues. Explore key switching techniques and MIDI CC (Continuous Controller) expression mapping for dynamic and expressive MIDI programming.	8	3
4	Practical Session		30	
	4.1	Editing Techniques: Teach students to edit MIDI performances by correcting notes, adjusting length (legato/staccato), fixing overlaps or duplicates, editing velocity, and using transpose and pitch shift tools.	8	4
	4.2	Quantization and MIDI-to-Audio Conversion: Practice applying quantization settings (input quantize, swing, humanize), and demonstrate converting MIDI to audio, exporting stems, freezing tracks, and preparing sessions for mixing.	7	4
	4.3	Mixing and Final Output: Guide students through importing stems, organizing track layouts, basic audio mixing techniques, and exporting the final mixdown for delivery or further production.	15	4

5	5.1	Teacher Specific Content		
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Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Theory Classes about the working of MIDI • Practical Sessions: Hands-on practice with using audio equipment. • Music production demonstration
Assessment Types	MODE OF ASSESSMENT Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) Practical Total Marks: 30 FA: Viva Voice, Assignment SA: Competency test in DAW
	B. End Semester Evaluation (ESE) Practical Total Marks: 70 , Duration: 3 hrs Project Evaluation: 60 marks Viva: 10 marks

Syllabus

References

1. Strange, Allen. *Electronic Music: Systems, Techniques, and Controls*. Dubuque, IA: Kendall Hunt Publishing, 1983.
2. Welsh, Fred. *Welsh's Synthesizer Cookbook: Synthesizer Programming, Sound Analysis, and Universal Patch Book*. Bradshaw Publishing, 2006.
3. Watkinson, John. *The Art of Digital Audio*. 3rd ed. Oxford: Focal Press, 2001.



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering				
Programme	MGU-B.Voc. (Honours) Sound Engineering				
Course Name	Sound Story				
Type of Course	SDC				
Course Code	MG4SDCSEG202				
Course Level	200				
Course Summary	This course introduces the art of storytelling through sound, develops creative skills in using voice, music, and sound effects, applies narrative structures and sound design techniques for immersive audio stories, and provides hands-on experience in creating original sound stories.				
Semester	4	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
	Lecture class, Hands-on Practicum sessions	3	1	0	75
Pre- requisites, if any	Nil				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the elements and structure of sound-based storytelling.	U	1,2,3,4,5, 6,7,8,9,10
2	Apply sound design principles to enhance narrative and emotional impact.	A	1,2,3,4,5, 6,7,8,9,10

3	Use voice, music, ambience, and effects effectively in storytelling.	A	1,2,3,4,5,6,7,8,9,10
4	Create, edit, and produce original sound stories using digital audio tools.	C	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	2	2	3	1	1	2	2	2	3
CO2	3	3	3	2	1	2	2	2	2	3
CO3	2	2	2	3	2	2	2	2	3	3
CO4	3	3	3	3	3	2	2	2	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Introduction to Sound Storytelling		15	
	1.1	Basics of storytelling: plot, characters, setting, conflict, resolution.	3	1
	1.2	Storytelling through sound: importance of audio in narratives. The role of listener imagination in sound stories.	7	1
	1.3	Types of sound stories: audio drama, podcast fiction, radio plays, narrative podcasts. Case studies: analysis of successful sound stories.	5	1

	Elements of Sound Story		15	
	2.1	Voice and narration techniques: tone, pacing, expression.	5	2
	2.2	Role of music: mood setting, transitions, emotional enhancement. Sound effects and ambience: creating space, movement, and realism.	3	2
	2.3	Balancing dialogue, sound effects, music, and silence. Scripting for sound stories.	7	2
3	Tools and Techniques for Sound Story Creation		15	
	3.1	Introduction to Digital Audio Workstations (DAWs). Microphones, recorders, and basic recording techniques.	5	3
	3.2	Editing and mixing: levels, EQ, panning, and effects. Sound libraries and creating original effects. Legal aspects: copyright, licensing, and ethical use of sound materials.	10	3
4	Practical Project - Sound Story Production		30	
	4.1	Concept development: choosing a story idea. Script writing and pre-production planning. Recording voice, music, and effects.	15	4
	4.2	Editing, mixing, and finalizing the sound story.	15	4

		Project presentation and peer reviews.		
5	5.1	Teacher Specific Content		

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Lectures, storytelling workshops, case studies, listening sessions. • Voice practice, script writing exercises, mood-building sessions, sound layering demos. • Studio demonstrations, hands-on editing sessions, software tutorials. • Each student/group produces a 3–5-minute original sound story demonstrating creative and technical proficiency
Assessment Types	MODE OF ASSESSMENT Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) Practical Total Marks: 30 FA: Viva Voice, Practical skills SA: Competency tests in DAW
	B. End Semester Evaluation (ESE) Practical Total Marks: 70 , Duration: 3 hrs Project Evaluation: 60 marks Viva Voice: 10 marks

References

1. McHugh, Siobhán & Lindsay, Richard. *The Power of Podcasting: Telling Stories Through Sound*. Routledge, 2022.
2. Sonnenschein, David. *Sound Design: The Expressive Power of Music, Voice and Sound Effects in Cinema*. Michael Wiese Productions, 2001.
3. Alten, Stanley R. *Audio in Media*. Cengage Learning, Latest Edition.
4. Beauchamp, Robin. *Designing Sound for Animation*. Focal Press, 2013.



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering			
Programme	MGU-B.Voc. (Honours) Sound Engineering			
Course Name	Sound for Broadcasting and Live Events			
Type of Course	SEC			
Course Code	MG4SECSEG200			
Course Level	200			
Course Summary	To develop theoretical knowledge of live sound reinforcement systems, broadcast audio workflows, and event planning with a focus on safety, communication, and legal responsibilities in professional environments.			
Semester	4	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
	Lecture class and demo sessions	3	0	0
Pre- requisites, if any	Nil			

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Explain the structure, signal flow, and theoretical design of live sound reinforcement systems.	U	1,2,3,4,5,6,7,8,9,10
2	Understand broadcast audio formats, standards, and practices across different media platforms.	U	1,2,3,4,5,6,7,8,9,10

3	Plan and evaluate the audio requirements for public events with attention to safety, team coordination, and compliance.	A	1,2,3,4,5,6,7,8,9,10
4	Recognize ethical, legal, and disaster-related factors relevant to professional sound operations in live and broadcast events.	U	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	3	2	2	1	1	1	1	2	3
CO2	3	3	3	2	1	1	1	1	2	3
CO3	3	3	3	2	3	3	2	2	3	3
CO4	2	2	2	2	2	3	2	3	2	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Live Sound & Broadcast Core Skills		15	
	1.1	PA system design basics (main speakers, delay fills, subwoofers) Stage monitors vs. in-ear monitoring systems Feedback control: mic placement, EQ notching, gating Role and workflow of a live mixing engineer	8	1
	1.2	Audio formats: mono, stereo, surround sound formats Broadcast types: studio, field (OB vans), sports, news, live telecasts	7	2

		International standards: ITU-R BS.1770, EBU R128 Loudness normalization: LKFS, LUFS and broadcast compliance		
	Event Planning, Safety & Management		15	
	2.1	Event Audio Planning: Interpreting event briefs and rider sheets Estimating speaker load and power requirements Venue-specific considerations (open ground vs. auditorium)	5	3
	2.2	Crew Management: Roles in an audio crew: FOH, monitor, stage tech, patch assistant Crew communication: intercom systems, cue signals Scheduling and rehearsal coordination	5	3
	2.3	Safety & Standards: Electrical safety, cabling, rigging Noise limits, temporary structure rules Introduction to international and Indian safety standards (OHS, NABH if applicable)	5	3
	Legal, Ethical, and Emergency Protocols		15	
3	3.1	Legal Requirements: Sound licensing, local authority permissions, insurance needs Copyrights and usage for background music, jingles, and live acts	5	4

	3.2	Ethical Practice: Respecting artist privacy, data protection in media broadcasts Inclusivity and accessibility at public events (subtitles, sign audio)	5	4
	3.3	Disaster & Contingency Management: Dealing with power failure, equipment malfunction, crowd panic Quick shutdown protocol, tech redundancy (backup systems) Case studies: successful vs. failed live/broadcast events	5	4
4	4.1	Teacher Specific Content		

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Concept-based lectures, audio signal flow diagrams • Team activity planning, safety demos, speaker setup maps, mock crew assignment • Case study analysis, group discussions, simulation of failure-response scenarios
Assessment Types	MODE OF ASSESSMENT Mode of Assessment

	A. Continuous Comprehensive Assessment (CCA) Practical Total Marks: 25 FA: Quiz, Assignment, Oral Presentations SA: Practical Test
	B. End Semester Evaluation (ESE) Practical Total Marks: 50, Duration: 2 hrs Lab Exam: 40 marks Viva Voice: 10 marks

References

1. Huber, David Miles, and Robert E. Runstein. *Modern Recording Techniques*. 9th ed. Oxford: Focal Press, 2018.
2. Rayburn, Ray A., ed. *Handbook for Sound Engineers*. 5th ed. New York: Focal Press, 2015.
3. Walne, Graham. *Live Sound: Theory and Practice*. London: Focal Press, 1991.
4. European Broadcasting Union (EBU) and International Telecommunication Union (ITU). *Loudness Recommendation R128 and BS.1770 Series Standards*. Geneva: ITU Publications, 2011–2020.
5. Audio Engineering Society (AES). *AES Broadcast and Streaming Audio Guidelines*. New York: AES Publications, various years.



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering				
Programme	MGU-B.Voc. (Honours) Sound Engineering				
Course Name	Electrical Aspects in Sound Engineering				
Type of Course	VAC				
Course Code	MG4VACSEG200				
Course Level	200				
Course Summary	This course introduces the basic concepts of electricity in simple, non-technical language, educates on the safe handling and operation of household electrical systems, creates awareness about safety devices, circuit breakers, and preventive measures, and promotes energy-saving practices along with early identification of electrical hazards.				
Semester	4	Credits		3	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
	Lecture class , practicum	3	0	0	45
Pre- requisites, if any	Nil				

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand basic electrical terms and household wiring systems.	U	1,2,3,4,5, 6,7,8,9,10
2	Identify different types of circuit breakers and their applications.	U	1,2,3,4,5, 6,7,8,9,10

3	Apply safety precautions and handle common household electrical issues.	A	1,2,3,4,5,6,7,8,9,10
4	Implement energy-saving techniques and detect early signs of electrical leakage.	S	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	2	3	2	1	1	2	2	2	1	3
CO2	2	3	2	1	1	2	2	2	1	3
CO3	3	3	2	2	2	3	3	3	2	3
CO4	3	3	2	2	2	3	3	3	2	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Basic Electrical Terminologies & Household Electrical Systems		15	
	1.1	What is electricity? Simple explanation of voltage, current, resistance, power. Alternating Current (AC) vs. Direct Current (DC). Introduction to household wiring systems: single-phase, three-phase, wiring colours.	3	1
	1.2	Common household electrical appliances and their power ratings. Understanding MCB (Miniature Circuit Breaker) box at home.	7	2

		How electricity reaches homes — basic understanding of power supply.		
	1.3	Types of circuit breakers: • Miniature Circuit Breakers (MCB) • Residual Current Circuit Breakers (RCCB) • Earth Leakage Circuit Breakers (ELCB) • Moulded Case Circuit Breakers (MCCB) Function and importance of each breaker. Safety devices: fuses, surge protectors, grounding systems. Practical identification of safety devices at home or workplace.	5	2
	Electrical Safety Parameters & Household Awareness		15	
	2.1	General safety rules when using electrical appliances. Safe use of extension cords, power strips, and portable devices. Avoiding overloading of circuits.	10	3
	2.2	Safe handling during repairs and wet conditions. Importance of professional servicing. First aid for electric shocks and burns. Reporting electrical issues to utility companies.	5	3

3	Energy Saving Methods & Earth Leakage Identification		15	
	3.1	Understanding energy consumption: reading electricity bills. Energy-saving appliances (star-rated equipment). Smart practices for energy conservation at home. Identifying faulty appliances causing high energy use.	5	4
	3.2	Earth leakage: • Symptoms of leakage (tingling sensation, minor shocks) • How RCCBs/ELCBs help prevent electrocution. • Professional inspection and periodic maintenance.	10	4
4	4.1	Teachers' Specific Content		

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) 1. Simple demonstrations, visual aids, real-life examples, case studies of common faults, videos, interaction with electricians 2. Interactive discussions, real-life examples, quizzes.
Assessment Types	MODE OF ASSESSMENT Mode of Assessment

	A. Continuous Comprehensive Assessment (CCA) Practical Total Marks: 25 FA: Quiz, Assignment, Oral Presentations SA: Practical Test
	B. End Semester Evaluation (ESE) Practical Total Marks: 50 , Duration: 2 hrs Lab Exam: 40 marks Viva Voice: 10 marks

References

1. Gupta, B. R. *Basic Electrical Engineering*. New Delhi: S. Chand Publishing, 2011.
2. Bureau of Energy Efficiency (BEE), Government of India. *Energy Conservation Guidebooks*. New Delhi: Ministry of Power, 2005–2018.
3. Institute of Electrical and Electronics Engineers (IEEE). *Public Safety Resources on Electrical Safety*. Piscataway, NJ: IEEE, 2010–2020.
4. Bureau of Indian Standards (BIS). *National Electrical Code: Code of Practice for Electrical Wiring Installations (IS 732)*. New Delhi: BIS, 2011.

MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering				
Programme	MGU-B.Voc. (Honours) Sound Engineering				
Course Name	Internship				
Type of Course	INT				
Course Code	MG4INTSEG200				
Course Level	200				
Course Summary	This course is designed to provide students with supervised industry experience in various fields of sound engineering such as recording studios, radio stations, post-production units, or live sound venues. It is conducted during Semester IV, typically over a minimum of 20 working days. The internship bridges academic knowledge and real-world practice, helping students develop workplace readiness, technical confidence, and professional communication. The course includes pre-internship orientation, mentor-based supervision, and post-internship evaluation through reports and presentations.				
Semester	4	Credits		2	Total Days
Course Details	Learning Approach	Lecture	Practical	OJT	
	Practical sessions	0	0	2	20
Pre-requisites, if any	Nil				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Identify the structure, workflow, and job roles in a professional sound engineering environment.	U	1,2,3,4,9,10
2	Apply basic sound engineering skills such as recording, editing, and monitoring in real-world tasks.	A	1,2,3,5,6,9,10

3	Demonstrate appropriate professional behaviour, time management, and team collaboration.	S	4,5,6,8,9,10
4	Prepare a reflective internship report documenting the learning experience, challenges, and achievements	C	1,3,4,6,9,10

Assessment Types	MODE OF ASSESSMENT	
	Mode of Assessment	
	A. Continuous Comprehensive Assessment (CCA)	
	Total Marks: 15	
	Components	Marks
	Feedback from the hosting organization	5
	Internal Supervisor Feedback	10
	B. End Semester Evaluation (ESE)	
	Total Marks: 35	
	Components	Marks
	Presentation	10
	Report	10
	Viva Voice	15

Syllabus



SEMESTER 5

MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering				
Programme	MGU-B.Voc. (Honours) Sound Engineering				
Course Name	New Media				
Type of Course	SDC				
Course Code	MG5SDCSEG300				
Course Level	300				
Course Summary	This course covers the evolution of new media, including social media, blogs, and podcasts. Students learn about digital content creation, audience engagement, and the impact of new media on society. It also covers tools, ethics, and future trends like AI and AR/VR. Practical skills in using digital media tools are developed.				
Semester	5	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
	Lecture Class , Demo Sessions	4	0	0	60
Pre- requisites, if any	Nil				

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the concept, evolution, and core characteristics of new media and how it differs from traditional forms.	U	1,2,3,4,5, 6,7,8,9,10

2	Analyse the social media shapes communication, journalism, public opinion, and online culture.	An	1,2,3,4,5,6,7,8,9,10
3	Students will gain hands-on exposure to content tools, understand media ethics, and explore upcoming technologies in the field.	U	1,2,3,4,5,6,7,8,9,10
4	Identify and apply legal and ethical standards in protecting copyright and ensuring responsible use of digital content.	A	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	2	3	2	1	2	1	2	1	2
CO 2	3	2	2	3	1	3	2	2	2	2
CO 3	2	3	3	3	2	2	1	2	2	3
CO 4	2	2	1	2	1	2	3	3	2	2

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Introduction to New Media		15	
	1.1	Evolution from traditional to new media, Characteristics of new media (interactivity, convergence, digitalization)	5	1
	1.2	Types of new media: websites, blogs, social media, OTT platforms, podcasts	5	1
	1.3	Role of the internet and mobile technologies, Impact of new media on journalism, education, and entertainment	5	1

2	Social media and Digital Communication		15	
	2.1	Major social media platforms and their purposes (Facebook, Instagram, X, YouTube)	4	2
	2.2	Digital content creation: text, audio, video, and live streams, Virality, memes, trends, and audience engagement	7	2
	2.3	Citizen journalism and user-generated content, Fake news, algorithms,	4	2
3	New Media Tools, Ethics, and Future Trends		15	
	3.1	Basic tools for digital content creation (Canva, Audacity, OBS, DAWs, CMS like WordPress)	6	3
	3.2	Legal and ethical issues in new media: privacy, copyright, trolling, cyberbullying	6	3
	3.3	Artificial Intelligence, AR/VR, and the future of media,	3	3
4	Open-source journalism		15	
	4.1	Responding to the audience, Annotative reporting, Citizen Journalists	6	4
	4.2	Problem of verification, accuracy and fairness	3	4
	4.3	Use of blogs, tweets, etc. for story generation and development, protecting copyright, Exploring Cyberspace:	6	4
5	5.1	Teacher specific Content		

Teaching and Learning Approach	Teaching Methods (Theory) • Interactive Lectures on the evolution and characteristics of new media • Case Studies of real-time social media campaigns and viral content • Film and Platform Analysis (YouTube, Instagram, OTT platforms) • Group Discussions on fake news, user-generated content, and ethics Learning Activities (Practical) • Hands-on Workshops on tools like Canva, Audacity, OBS, WordPress • Digital Content Creation Tasks: Create blog posts, videos, or podcasts • Mock Campaigns on audience engagement and virality • Analysis Tasks: Track social media trends and algorithms							
Assessment Types	MODE OF ASSESSMENT Mode of Assessment							
	A. Continuous Comprehensive Assessment (CCA) Theory Total Marks: 30 FA: Quiz, Oral Presentations SA: Written Test							
	B. End Semester Evaluation (ESE) Theory Total Marks: 70, Duration: 2 hrs <table border="1"> <thead> <tr> <th>Components</th><th>Marks</th></tr> </thead> <tbody> <tr> <td>Very short answer</td><td>10 out of 12 – 10 x 2 = 20 marks</td></tr> <tr> <td>Short answer</td><td>6 out of 9 – 6 x 5 = 30 marks</td></tr> <tr> <td>Essay</td><td>2 out of 3 – 10 x 2 = 20 marks</td></tr> </tbody> </table>	Components	Marks	Very short answer	10 out of 12 – 10 x 2 = 20 marks	Short answer	6 out of 9 – 6 x 5 = 30 marks	Essay
Components	Marks							
Very short answer	10 out of 12 – 10 x 2 = 20 marks							
Short answer	6 out of 9 – 6 x 5 = 30 marks							
Essay	2 out of 3 – 10 x 2 = 20 marks							

References

1. Siapera, Eugenia. *Understanding New Media*. London: SAGE Publications, 2012.
2. Crosbie, Vin. "What Is New Media?" *Converge Digital Journalism*. Accessed July 12, 2025.
3. Baran, Stanley J. *Introduction to Mass Communication: Media Literacy and Culture*. 9th ed. New York: McGraw-Hill Education, 2014.
4. Flew, Terry. *New Media: An Introduction*. 3rd ed. Oxford: Oxford University Press, 2014.



MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering			
Programme	MGU-B.Voc. (Honours) Sound Engineering			
Course Name	Studio Acoustics			
Type of Course	SDE			
Course Code	MG5SDESEG300			
Course Level	300			
Course Summary	This course teaches the principles of studio design and acoustic treatment. Students will explore different studio types, the difference between acoustics and soundproofing, and how to control room reflections and reverberation. The course also covers key acoustic materials, studio layout considerations, and methods for measuring and optimizing room acoustics.			
Semester	5	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
	Lecture class, Hands-on Practicum sessions	3	1	0
Pre- requisites, if any	Nil			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Differentiate between studio types and evaluate their design, functionality, and suitability for various production needs.	U	1,2,3,4,6, 7,8,9,10

2	Understand the principles of acoustics and soundproofing, including appropriate material selection for each purpose.	U	1,2,3,4,5,6,7,8,9,10
3	Design acoustically optimized studio spaces by addressing room symmetry, isolation, and treatment with diffusers, absorbers, and bass traps	A	1,2,3,4,5,6,7,8,9,10
4	Measure and analyze room acoustics using measurement tools and software for both symmetrical and asymmetrical spaces.	A	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	3	2	1	0	1	1	1	1	2
CO2	2	3	2	1	1	1	1	1	1	2
CO3	2	3	2	2	1	1	1	1	2	2
CO4	2	3	2	1	2	1	1	1	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Studio Types		15	
	1.1	Explore Studio Types – Understand the characteristics of professional, project, and home studios.	6	1
	1.2	Compare Advantages and Limitations – Evaluate the strengths and weaknesses of each studio type.	4	1
	1.3	Identify Key Differentiators – Learn what makes one studio type stand out over others in various production contexts.	5	1

2	Acoustics and Sound Proofing		15	
	2.1	Technical aspects of sound regarding acoustic design	5	2
	2.2	Advantages and disadvantages of each studio type	5	1
	2.3	Factors affecting the design of a studio	5	2
3	Studio Design		15	
	3.1	Key acoustic principles for studio and control room design, including isolation and symmetry. Managing room reflections and reverberation using diffusers, bass traps, and acoustic chambers	10	3
	3.2	Practical applications of acoustic treatments to optimize sound quality in studio spaces.	5	4

4	Practical - Measurements and Calculations		30	
	4.1	Using measurement microphones and software to assess room acoustics.	8	4
	4.2	Techniques for measuring symmetrical and asymmetrical rooms accurately.	10	4
	4.3	Applying measurement results to optimize studio acoustic design.	12	4
5	5.1	Teacher Specific Content		

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Real-life examples to explain acoustics • Live demonstrations of acoustic designing scenarios • Hands-on practice by designing a room
Assessment Types	MODE OF ASSESSMENT Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) Practical Total Marks: 30 FA: Viva Voice, Assignments, Oral Presentation SA: Practical test
	B. End Semester Evaluation (ESE) Practical Total Marks: 70 , Duration: 3 hrs Lab Exam: 60 marks Viva: 10 marks

References

1. Everest, F. Alton, and Ken C. Pohlmann. *Master Handbook of Acoustics*. 6th ed. New York: McGraw-Hill Education, 2014.
2. Gallagher, Mitch. *Acoustic Design for the Home Studio*. Milwaukee: Hal Leonard Corporation, 2012.
3. Everest, F. Alton. *Sound Studio Construction on a Budget*. 2nd ed. New York: McGraw-Hill Education, 2001.



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering				
Programme	MGU-B.Voc. (Honours) Sound Engineering				
Course Name	Stereo Amplifier Assembling and Testing				
Type of Course	SDE				
Course Code	MG5SDESEG301				
Course Level	300				
Course Summary	To understand the theoretical principles behind amplifier design and functioning, study the components involved in amplifier circuits, gain hands-on experience in assembling, testing, and troubleshooting stereo amplifiers, and learn safety protocols and industry standards in electronics assembly.				
Semester	5	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
	Lecture class, Hands-on Practicum sessions	3	1	0	75
Pre- requisites, if any	Nil				

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the theoretical principles, types, and design blocks of audio amplifiers.	U	1,2,3,4,6, 7,8,9,10

2	Analyse the function of components in amplifier circuits and power supplies.	An	1,2,3,4,5,6,7,8,9,10
3	Apply systematic methods for soldering, circuit assembly, testing, and fault diagnosis.	A	1,2,3,4,5,6,7,8,9,10
4	Assemble and test a fully functional stereo amplifier following safety and quality protocols.	A	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	3	2	1	1	1	1	1	2	3
CO2	3	3	3	1	1	1	1	1	2	3
CO3	3	3	3	2	2	1	1	2	3	3
CO4	3	3	3	2	2	2	2	2	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Fundamentals of Audio Amplifiers		15	
	1.1	Introduction to amplifiers: purpose and audio applications	6	1
	1.2	Classification: Class A, B, AB, D amplifiers. Efficiency, distortion, thermal considerations	4	1
	1.3	Block diagram of stereo amplifier system. Heat dissipation and protection circuit basics.	5	1
2	Components and Circuit Analysis		15	
	2.1	Active & passive components: resistors, capacitors, transistors, ICs	5	2

	2.2	Power supply block: transformer, rectifier, voltage regulation	5	1
	2.3	Circuit protection: fuses, relays, current limiting. PCB layout basics and schematic reading	5	2
	Assembly Process and Troubleshooting		15	
3	3.1	Tools: soldering station, multimeter, oscilloscope. Soldering techniques, error corrections	10	3
	3.2	Assembly procedures, biasing and calibration. Diagnosing faults: hum, distortion, short circuits	5	3

	Practical: Stereo Amplifier Assembling and Testing		30	
4	4.1	Component testing & PCB preparation. Power supply verification	6	3
	4.2	Full assembly of stereo amplifier. Final testing: gain, frequency response, thermal monitoring	15	3
	4.3	Fault simulation and troubleshooting practice.	9	4
5	5.1	Teacher Specific Content		

Syllabus

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) - Understand how amplifiers work and the basic design goals for high-quality stereo amplifiers. - Analyze amplifier circuits and identify the function of various components. - Learn systematic approaches to assembly, testing, and fault diagnosis. - Assemble and test a complete stereo amplifier, gaining hands-on experience.
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Assessment Types	MODE OF ASSESSMENT
	Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) Practical Total Marks: 30 FA: Viva Voice, Assignments SA: Practical test B. End Semester Evaluation (ESE) Practical Total Marks: 70, Duration: 3 hrs Project Evaluation: 60 marks Viva Voice: 10 marks

References

1. Horowitz, Paul, and Winfield Hill. *The Art of Electronics*. 3rd ed. Cambridge: Cambridge University Press, 2015.
2. Rosenblit, Elliot. *Audio Amplifier Projects*. New York: Elektor International Media, 2012.
3. Smith, Douglas. *High-Power Audio Amplifier Construction Manual*. New York: McGraw-Hill, 1997.
4. Gibson, Bill. *The AudioPro Home Recording Course: A Complete Course in the Music Production Process*. Boston: Course Technology PTR, 2007.
5. Miller, Rex. *Electronics: A Self-Teaching Guide*. 2nd ed. Hoboken, NJ: Wiley, 2004.



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering			
Programme	MGU-B.Voc. (Honours) Sound Engineering			
Course Name	Multitrack Audio Production			
Type of Course	SDC			
Course Code	MG5SDCSEG301			
Course Level	300			
Course Summary	To develop the skills needed for music production by covering key areas of pre-production planning, professional recording techniques, digital mixing processes, and stereo mastering through hands-on and theoretical instruction.			
Semester	5	Credits		Total Hours
Course Details	Learning Approach Lecture class, Hands-on Practicum sessions	Lecture	Practical	
		3	1	0
Pre- requisites, if any	Nil			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Plan the pre-production process including song arrangement, artist coordination, and technical documentation.	A	1,2,3,4,5,6,7,8,9,10
2	Set up and operate microphones and DAW sessions for professional multitrack audio recording.	A	1,2,3,4,5,6,7,8,9,10

3	Apply techniques of mixing such as session organization, EQ, dynamics processing, and automation.	A	1,2,3,4,5,6,7,8,9,10
4	Execute a complete stereo music production project, including mixdown and mastering for final output.	C	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	2	2	2	2	1	1	1	2	3
CO2	3	3	2	1	2	1	1	1	3	3
CO3	3	3	3	2	2	1	1	1	3	3
CO4	3	3	3	2	3	2	1	1	3	3

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COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Pre-production & Session Setup		15	
	1.1	Song selection, structure planning, and artist arrangement Creating detailed log sheets, music charts, notes	3	1
	1.2	Planning recording order of instruments and parts Introduction to mono vs stereo mic techniques	7	1
	1.3	Selection of appropriate microphones for specific sound DAW project setup: track layout, headphone mixes, monitoring chains	5	1

	Recording Techniques		15	
	2.1	Mono vs. stereo mic techniques for different instruments Choosing microphones: dynamic, condenser, ribbon, USB	5	2
	2.2	Setting up audio interfaces and DAW templates	3	2
	2.3	Headphone mixes, monitoring setup, recording levels Multitrack vocal and instrument recording sessions	7	2
3	Mixing Concepts		15	
	3.1	Arranging sessions: labelling, grouping, bussing Editing clips, fades, trims, time alignment	5	3
	3.2	Balancing tracks, applying EQ, compression, reverb Automation of volume, pan, effects Basic mastering: loudness normalization, export formats	10	3
4	Final Stereo Production		30	
	4.1	Project: Record a full-length song using live and/or electronic instruments	10	4
	4.2	Perform multitrack editing, mixing, and mastering	10	4
	4.3	Export as stereo WAV/MP3 file in streaming-ready format	10	4

		Submit session files with documentation		
5	5.1	Teacher Specific Content		

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Lecture, discussion, project planning sheets • Signal flow diagrams, mic placement visuals • Audio demos, session analysis, plugin walkthroughs • Assessment Criteria: Clarity, balance, creativity, technical accuracy
Assessment Types	MODE OF ASSESSMENT Mode of Assessment:
	A. Continuous Comprehensive Assessment (CCA) Practical Total Marks: 30 FA: Viva Voice, Assignments SA: Competency Tests in DAW
	B. End Semester Evaluation (ESE) Practical Total Marks: 70 , Duration: 3 hrs Project Evaluation: 50 marks Report: 10 marks Viva Voice: 10 marks

References

1. Huber, David Miles, and Robert E. Runstein. *Modern Recording Techniques*. 9th ed. New York: Focal Press, 2017.
2. Owsinski, Bobby. *The Mixing Engineer's Handbook*. 5th ed. Milwaukee: Hal Leonard, 2017.
3. Senior, Mike. *Mixing Secrets for the Small Studio*. 2nd ed. New York: Focal Press, 2014.
4. Izhaki, Roey. *Mixing Audio: Concepts, Practices, and Tools*. 3rd ed. New York: Focal Press, 2018.



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering				
Programme	MGU-B.Voc. (Honours) Sound Engineering				
Course Name	Art and Technology of Mixing and Mastering				
Type of Course	SEC				
Course Code	MG5SECSEG300				
Course Level	300				
Course Summary	This course aims to develop advanced skills in mixing audio with balance, clarity, and creativity, introduce key mixing tools such as compression, EQ, dynamic processing, and time-based effects, train students in advanced FX usage, automation techniques, and hybrid mixing systems, and equip them with essential mastering concepts for professional release formats.				
Semester	5	Credits		3	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
	Practicum	2	1	0	
Pre- requisites, if any	Nil				

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Explain the foundational principles, tools, and workflow involved in mixing and mastering.	U	1,2,3,4,5, 6,7,8,9,10
2	Apply techniques of mix preparation, balancing, dynamics, and spatial effects using digital and hybrid setups.	A	1,2,3,4,5, 6,7,8,9,10

3	Use advanced signal routing, automation, and analog-digital integration within a DAW environment.	A	1,2,3,4,5,6,7,8,9,10
4	Perform a complete mixing and mastering workflow on a 2-track or multitrack session suitable for professional distribution.	A	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	3	2	2	1	1	1	1	2	3
CO2	3	3	3	2	2	1	1	1	3	3
CO3	3	3	3	2	2	1	1	1	3	3
CO4	3	3	3	2	3	2	2	2	3	3

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COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Theoretical Foundations of Mixing		15	
	1.1	Mixing overview: "Tall, deep, and wide" approach Elements of a mix: Balance, frequency range, panorama, dynamics, interest Monitoring hardware: Studio monitors, headphones, subwoofers	7	1
	1.2	Room acoustics and ear fatigue Mixing environments: ITB (in-the-box) vs OTB (out-of-the-box)	3	1

	1.3	Analog signal flow, hybrid mixing chains Serial vs. parallel processing Reference listening and evaluation	5	1
	Creative Mix Techniques and Mastering Concepts		15	
	2.1	Mix Preparation: Track organization, markers, color coding, fades Tuning and timing corrections (Autotune, Melodyne) Routing: Buses, aux, FX, VCAs	4	2
	2.2	Balancing and Dynamics: Gain staging, metering, compression, EQ types Ducking, de-essing, multiband compression Saturation and harmonic enhancers	4	2
	2.3	FX and Automation: Reverb (types, convolution, IR), Delay (slapback, stereo) Stereo widening, analog gear integration Automation modes and tricks Mastering: Need and process of mastering	7	3

		Loudness standards (LUFS, RMS) Stem mastering, final formats, streaming compliance		
3	Practical Mixing & Mastering Project		30	
	3.1	Students mix and master a full-length multitrack session Include effects routing, automation, and export to industry standard	20	4
	3.2	Submission of before/after mix samples Evaluation based on clarity, loudness, balance, and creativity	10	4
4	4.1	Teachers' Specific Content		

MGU-B.VOC. (HONOURS)

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) 1. Theory lectures, signal flow diagrams, listening comparisons 2. DAW session walkthroughs, plugin analysis, video case studies 3. Assessment Parameters: Creativity, clarity, technical execution, loudness accuracy
Assessment Types	MODE OF ASSESSMENT Mode of Assessment

	A. Continuous Comprehensive Assessment (CCA) Practical Total Marks: 25 FA: Viva Voice, Practical Assignment SA: Competency tests in DAW
	C. End Semester Evaluation (ESE) Practical Total Marks: 50, Duration: 3 hrs Lab Exam: 40 marks Viva: 10 marks

References

1. Katz, Bob. *Mastering Audio: The Art and the Science*. 3rd ed. New York: Focal Press, 2015.
2. Owsinski, Bobby. *The Mixing Engineer's Handbook*. 5th ed. Milwaukee: Hal Leonard, 2017.
3. Senior, Mike. *Mixing Secrets for the Small Studio*. 2nd ed. New York: Focal Press, 2014.
4. Huber, David Miles, and Robert E. Runstein. *Modern Recording Techniques*. 9th ed. New York: Focal Press, 2017.



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering			
Programme	MGU-B.Voc. (Honours) Sound Engineering			
Course Name	Sonic Awareness and Public Responsibility			
Type of Course	VAC			
Course Code	MG5VACSEG300			
Course Level	300			
Course Summary	This course introduces the basics of sound and its effects, raises awareness about sound pollution and its social impact, educates on government guidelines and safety standards, and promotes individual responsibility in maintaining a healthy sonic environment.			
Semester	5	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
	Lecture class and demo sessions	3	0	0
Pre- requisites, if any	Nil			

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Explain basic sound concepts, including volume, pitch, decibels, and hearing range in a simplified manner.	U	1,2,3,4,5,6,7,8,9,10
2	Identify the causes, types, and harmful effects of sound pollution in public and domestic settings.	U	1,2,3,4,5,6,7,8,9,10

3	Understand government rules and permissible sound levels, especially in silence zones and public spaces.	U	1,2,3,4,5, 6,7,8,9,10
4	Promote responsible and ethical sound practices in the community through awareness and personal behavior.	A	1,2,3,4,5, 6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	2	2	2	1	1	1	1	2	2
CO2	3	3	3	2	1	2	2	1	2	3
CO3	2	3	3	2	1	3	2	2	2	3
CO4	2	2	2	3	2	3	3	3	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Understanding Sound – The Basics		15	
	1.1	What is sound? Frequency, amplitude, pitch (explained in simple terms) How we hear – the human ear and safe hearing range	3	1
	1.2	Decibel (dB) scale and volume awareness Examples of common sounds: nature, music, machines, traffic	7	1
	1.3	Listening activity: Safe vs. unsafe sound levels Sound in Indian traditional healing and rituals (optional contextual content)	5	1

	Sound Pollution and Public Impact		15	
	2.1	Definition of sound pollution Major sources: traffic, construction, industry, household appliances, festivals	5	2
	2.2	Effects of excessive noise on health: stress, sleep, hearing loss Special focus: Impact on children, elderly, and animals	3	2
	2.3	Sound pollution in Kerala: local case studies Difference between sound and noise	7	2
	Sound Responsibility & Government Guidelines		15	
	3.1	CPCB and MoEFCC noise pollution rules Permissible noise levels for residential, commercial, industrial, and silence zones	5	3,4
3	3.2	Penalties for violations, legal rights of citizens Tools for measuring noise: mobile apps, sound meters Role of individuals: turning off horns, lowering speaker volumes, mindful listening Community awareness campaigns and DIY posters	10	3,4
4	4.1	Teacher Specific Content		

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) 1. Interactive lectures, audio demos, group listening tasks 2. street audio recordings, group reflections, infographics 3. Policy review sessions, practical demo of apps/meters, group awareness poster/project
Assessment Types	MODE OF ASSESSMENT Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) Theory Total Marks: 25 FA: Quiz, Assignment, Oral Presentations SA: Written Test
	B. End Semester Evaluation (ESE) Theory Total Marks: 50 , Duration: 1 hr MCQ Exam

MGU-B.VOC. (HONOURS)

References

1. Central Pollution Control Board. *Noise Pollution (Regulation and Control) Rules, 2000*. Ministry of Environment, Forest and Climate Change, Government of India.
2. World Health Organization. *Environmental Noise Guidelines for the European Region*. WHO Regional Office for Europe, 2018.
3. Kerala State Biodiversity Board. *Soundscape and Acoustic Ecology Studies in Kerala*. Thiruvananthapuram, KSBB Publications.
4. Ministry of Environment, Forest and Climate Change. *Noise Standards and Control Measures in India*. Government of India.

SEMESTER 6



MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering				
Programme	MGU-B.Voc. (Honours) Sound Engineering				
Course Name	Dubbing and Sync Sound				
Type of Course	SDE				
Course Code	MG6SDESEG300				
Course Level	300				
Course Summary	This course introduces dubbing, voice-over, and expressive vocal performance techniques, trains students on sync sound recording and field production equipment, develops skills in Foley editing, sound design, and creative audio for visuals, and provides hands-on experience in full post-production workflows from pre-mix to final mix.				
Semester	6	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
	Practicum	3	1	0	
Pre- requisites, if any	Nil				

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Demonstrate voice dubbing and modulation techniques for narration and commentary.	A	1,2,3,4,5, 6,7,8,9,10
2	Operate field recording equipment and apply sync sound editing principles.	S	1,2,3,4,5, 6,7,8,9,10

3	Apply Foley editing and creative sound design techniques for video content.	A	1,2,3,4,5,6,7,8,9,10
4	Perform complete post-production mixing, mastering, and final delivery for audio-visual projects.	S	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	3	2	1	1	2	1	1	2	2
CO2	3	3	3	2	2	2	1	1	2	3
CO3	2	3	2	1	1	2	2	1	1	2
CO4	3	3	3	2	2	1	1	1	3	3

‘0’ is No Correlation, ‘1’ is Slight Correlation (Low level), ‘2’ is Moderate Correlation (Medium level) and ‘3’ is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Voice Dubbing & Narration		15	
	1.1	Dubbing Techniques: Narration, Commentary, and Dialogue Replacement. Voice Modulation Techniques: Tone, Pitch, Emotion, Expressiveness	5	1
	1.2	Microphone Technique for Voice Recording. Studio Setup for Dubbing	5	1
	1.3	Voice Acting & Script Interpretation. Common Challenges in Dubbing. Synchronization with Picture	5	1

2	Sync Sound & Location Recording		15	
	2.1	Sync Sound Basics: Location Recorders (Field Recorders, Record Formats). Location Microphones – Shotgun, Lavalier, Boundary Mics. Field Mixer Usage and Setup	7	2
	2.2	Boom Operation Techniques. Sound Editing Techniques: Effects & Ambience Recording. Track Laying of Location Sound. Dialogue Clean-up and Noise Reduction	5	2
	2.3	Basic ADR (Automated Dialogue Replacement)	3	2
3	Foley Editing & Sound Design		15	
	3.1	Foley Recording Techniques: Footsteps, Props, Clothing Rustle, Room Tones. Foley Libraries, Sample Packs & Sound Assets	5	3
	3.2	Ambience & Environmental Sound Layering. Introduction to Sound Design: Sound Designing for Visuals. Creative Layering & Combining Sound. Sound Design using Synthesizers, Samplers, Instruments	4	3
	3.3	Syncing Foley & Designed Sounds to Picture	6	3
4	Practical - Audio Post Workflow		30	
	4.1	Dialogue Pre-mix, Effects & Foley Premix, Ambience Premix, Music Premix. Final Mixing	15	4

	4.2	Project File Setup in DAW. Mixing Formats: Mono, Stereo and Surround. Dialog & Effects Stems Creation Output Formats for Broadcast/Streaming. Sync with Video, Export and Delivery	15	4
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Teaching and Learning Approach	• Concept-Oriented Lectures Introduce dubbing, voice-over, sync sound, Foley, and sound design through interactive lectures, video examples, and breakdowns of real-world audio post-production projects. • Demonstrations & Studio Walkthroughs Demonstrate the use of field recorders, boom microphones, dubbing booths, and Foley stages to familiarize students with industry-standard tools and workflows. • Practical Lab Sessions Provide hands-on training in voice recording, Foley creation, sound editing, and multitrack mixing using professional DAWs and audio post-production environments. • Project-Based Learning Guide students in end-to-end audio post-production projects that involve dubbing, sync sound, Foley, effects, premixing, final mix, and video output for evaluation.
Assessment Types	MODE OF ASSESSMENT
	Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) Practical Total Marks: 30 FA: Viva Voice, Practical Assignment SA: Observation of practical skills, Competency tests in DAW B. End Semester Evaluation (ESE) Practical Total Marks: 70, Duration: 3 hrs Lab Exam: 60 marks Viva: 10 marks

References

1. Yewdall, David Lewis. *Practical Art of Motion Picture Sound*. 4th ed. New York: Focal Press, 2011.
2. Holman, Tomlinson. *Sound for Film and Television*. 3rd ed. New York: Focal Press, 2010.
3. Sonnenschein, David. *Sound Design: The Expressive Power of Music, Voice and Sound Effects in Cinema*. Studio City, CA: Michael Wiese Productions, 2001.
4. Rose, Jay. *Producing Great Sound for Film and Video*. 4th ed. New York: Focal Press, 2014.



MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering				
Programme	MGU-B.Voc. (Honours) Sound Engineering				
Course Name	Audio Remastering Techniques				
Type of Course	SDE				
Course Code	MG6SDESEG301				
Course Level	300				
Course Summary	To provide an in-depth understanding of the concepts, tools, and aesthetics of audio remastering, covering signal restoration, loudness optimization, spectral correction, and delivery standards while equipping students with hands-on skills in remastering vintage and modern audio using DAW environments.				
Semester	6	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
	Practicum	3	1	0	
Pre- requisites, if any	Nil				

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Explain the purpose, history, and processes of audio remastering.	U	1,2,3,4,6, 9,10
2	Identify and apply signal processing tools to restore and enhance audio.	A	1,2,3,5,6, 8,9,10
3	Implement industry standards for mastering loudness, formats, and metadata.	A	1,2,3,4,7, 8,9,10

4	Perform a complete remastering workflow on legacy or mixed audio content.	S	1,2,3,4,5,6,7,8,9,10
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CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	3	2	2	1	2	1	1	2	3
CO2	3	3	3	1	2	2	1	2	3	3
CO3	3	3	3	3	1	2	2	2	3	3
CO4	3	3	3	3	3	3	2	2	3	3

‘0’ is No Correlation, ‘1’ is Slight Correlation (Low level), ‘2’ is Moderate Correlation (Medium level) and ‘3’ is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Fundamentals of Audio Remastering		15	
	1.1	What is remastering? Differences from mixing and original mastering. History and evolution of remastering.	5	1
	1.2	Goals: fidelity, dynamic range, clarity, loudness balance	5	1
	1.3	Understanding legacy formats: tape, vinyl, early digital. Types of remastering: stereo, multi-track, upmixing.	5	1

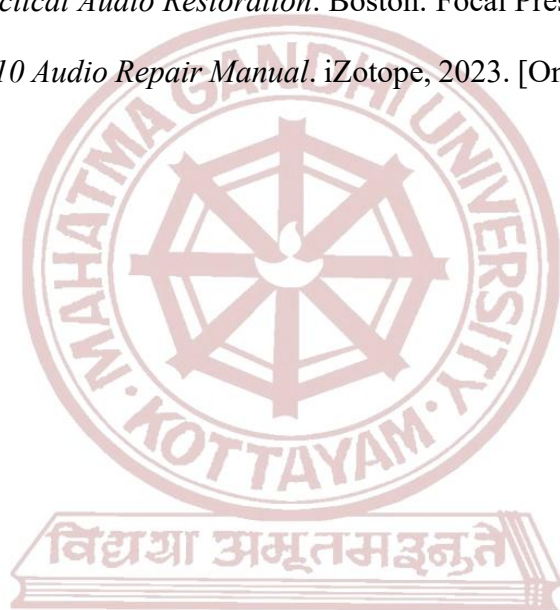
2	Tools and Techniques		15	
	2.1	Equalization strategies for remastering. Dynamic range compression and limiting.	7	2
	2.2	De-noising and de-clicking techniques. Use of spectral editors (iZotope RX, SpectraLayers)	5	2
	2.3	Audio restoration: hum, hiss, wow/flutter, distortion. Stereo image enhancement and phase correction.	3	2
3	Industry Standards & Delivery		15	
	3.1	Loudness standards: LUFS, True Peak, RMS	5	3
	3.2	Target formats: CD, streaming, vinyl, archival. Bit depth, sample rate, dithering, file formats.	4	3
	3.3	Metadata tagging, ISRC, embedding artwork. Case studies: remastered albums & restoration ethics	6	
4	Practical: Remastering Workflow		30	
	4.1	Importing audio and reference session prep. Noise profiling and removal Corrective EQ and balancing Level matching and loudness normalization	15	4

		Applying analog-style saturation and harmonic enhancement.		
	4.2	Exporting deliverables for various platforms. Final project: remaster a provided track to industry-ready quality.	15	4

Teaching and Learning Approach	• Blended Instructional Methods: Use a mix of classroom lectures, audio demonstrations, and visual walkthroughs to explain theoretical concepts like restoration, loudness standards, and legacy format handling. • Software-Based Practice: Hands-on sessions with industry-standard tools such as iZotope RX, FabFilter, Ozone, and DAWs like Pro Tools or Reaper to reinforce signal processing and remastering techniques. • Case-Based Learning: Analyze real-world remastered tracks to connect technical skills with aesthetic judgment and industry standards. • Project-Based Assessment: Final practical task includes remastering a degraded or vintage audio file with logbook documentation, peer review, and output in multiple formats (CD, streaming, archival).
Assessment Types	MODE OF ASSESSMENT Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) Practical Total Marks: 30 FA: Viva Voice, Practical Assignment SA: Observation of practical skills, Competency tests in DAW
	B. End Semester Evaluation (ESE) Practical Total Marks: 70 , Duration: 3 hrs Lab Exam: 60 marks Viva: 10 marks

References

1. Katz, Bob. *Mastering Audio: The Art and the Science*. 3rd ed. Burlington, MA: Focal Press, 2014.
2. Case, Alexander U. *Mix Smart: Pro Audio Tips for Your Multitrack Mix*. 2nd ed. New York: Focal Press, 2016.
3. Senior, Mike. *Mixing Secrets for the Small Studio*. 2nd ed. Oxford: Focal Press, 2017.
4. Watkinson, John. *The Art of Digital Audio*. 3rd ed. Oxford: Focal Press, 2013.
5. Gilder, Eric. *Practical Audio Restoration*. Boston: Focal Press, 2008.
6. iZotope Inc. *RX 10 Audio Repair Manual*. iZotope, 2023. [Online manual and tutorials]



MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering				
Programme	MGU-B.Voc. (Honours) Sound Engineering				
Course Name	Surround Sound Production				
Type of Course	SDC				
Course Code	MG6SDCSEG300				
Course Level	300				
Course Summary	This course helps students understand the history, formats, and concepts of multichannel surround sound, learn speaker configuration, calibration, and room acoustics for surround playback, explore various surround sound recording and immersive audio techniques, and develop hands-on skills for mixing surround sound content using DAWs				
Semester	6	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
	Practicum	3	1	0	
Pre- requisites, if any	Nil				

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Explain the principles and evolution of surround sound formats and systems.	U	1,2,3,4,5,6,7,8,9,10
2	Configure speaker systems and calibrate setups for accurate surround monitoring.	A	1,2,3,4,5,6,7,8,9,10

3	Apply multichannel recording techniques, including binaural and ambisonics.	A	1,2,3,4,5,6,7,8,9,10
4	Create complete surround sound mixes for audio-visual content using DAW tools.	C	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	3	2	1	1	1	1	1	3	3
CO2	3	3	2	2	1	2	2	1	3	3
CO3	3	3	3	2	2	2	2	1	2	3
CO4	3	3	3	2	2	2	2	1	2	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Introduction to Surround Sound		15	
	1.1	Basics of Multichannel Audio: Mono vs Stereo vs Surround. Historical Development of Surround Sound	3	1
	1.2	Commercial Surround Sound Formats: Dolby Surround, Dolby Digital, Dolby Atmos. DTS, DTS-HD, Auro 3D, Ambisonics. Home Theatre Formats and Layouts	7	1
	1.3	Object-Based vs Channel Based Audio	5	1
	Surround System Setup & Calibration		15	
	2.1	Speaker Layouts: 5.1, 6.1, 7.1, LCR systems, Immersive	6	2

		Formats.Understanding LFE/Subwoofer Channels		
	2.2	Speaker Placement & Acoustic Treatment for Surround Rooms Speaker Calibration Methods	5	2
	2.3	Home Theatre Setup, Devices and Compatibility Object-Based Audio Setup: Dolby Atmos Beds & Objects	4	2
	Surround Mixing in Protools		15	
3	3.1	Importing video, Preparing the session, create busses, multichannel setup, basic surround panning techniques	5	3
	3.2	Placing dialog and effects in the mix, automation, creating final mix downs.	10	3
	Practical - Surround Mixing		30	
4	4.1	DAW Setup for Surround Mixing (Pro Tools Focused). I/O Setup, Multichannel Bus Routing, Track Output Configuration. Surround Panning Techniques, Panning Modes. Importing Video for Audio Post Projects	15	4
	4.2	Multichannel Plugins & Effects. Surround Automation Techniques Up mixing and Down mixing Processes. Preparing Final Mix downs for Delivery	15	4
5	5.1	Teacher Specific Content		

Teaching and Learning Approach	• Concept-Based Lectures Explain surround sound formats, signal flow, speaker layouts, and immersive audio concepts using diagrams, multimedia, and historical case studies. • Demonstrations and Studio Walkthroughs Demonstrate surround monitoring setups, speaker calibration tools, room acoustics adjustments, and software-based routing in lab environments. • Practical Recording and Mixing Exercises Guide students through hands-on tasks involving surround mic techniques, Ambisonic recordings, binaural setups, and multichannel DAW configurations. • Project-Based Learning Students complete a surround sound mix project for video or VR, applying learned techniques in recording, panning, effects, and final output formatting.
Assessment Types	MODE OF ASSESSMENT Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) Practical Total Marks: 30 FA: Viva Voice, Practical Assignment SA: Observation of practical skills, Competency tests in DAW
	B. End Semester Evaluation (ESE) Practical Total Marks: 70 , Duration: 3 hrs Lab Exam: 60 marks Viva: 10 marks

References

1. Holman, Tomlinson. *Surround Sound: Up and Running*. 2nd ed. New York: Focal Press, 2007.
2. Rumsey, Francis. *Spatial Audio*. 2nd ed. New York: Focal Press, 2012.
3. Izhaki, Roey. *Mixing Audio: Concepts, Practices and Tools*. 3rd ed. New York: Focal Press, 2018.
4. Sonnenschein, David. *Sound Design: The Expressive Power of Music, Voice and Sound Effects in Cinema*. Studio City, CA: Michael Wiese Productions, 2001.



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering				
Programme	MGU-B.Voc. (Honours) Sound Engineering				
Course Name	Principles and Techniques of Radio Production				
Type of Course	SEC				
Course Code	MG6SECSEG300				
Course Level	300				
Course Summary	This course introduces students to audio communication theories, radio program formats, scriptwriting, sound design, and production analysis to help them create effective and engaging radio content.				
Semester	6	Credits		3	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
	Lecture Class , demo sessions	3	0	0	45
Pre- requisites, if any	Nil MGU-B.VOC. (HONOURS)				

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand basic radio formats and theories, and learn about radio ethics, copyright rules, and announcer conduct.	U	1,2,3,4,5, 6,7,8,9,1 0

2	Identify and analyze various radio program formats and structures, such as news bulletins, talk shows, radio dramas, music programs, documentaries, PSAs, and phone-in show, focusing on their purpose, style, and audience engagement.	An	1,2,3,4,5,6,7,8,9,10
3	Demonstrate knowledge of scriptwriting and sound design principles in radio production by exploring the use of voice modulation, sound effects, storytelling techniques, pacing, and flow in creating effective audio content.	A	1,2,3,4,5,6,7,8,9,10
4	Create radio production from a historical, psychological, and communicative perspective, understanding the evolution of production techniques, audience behavior, and the application of communication models like Shannon-Weaver in content creation.	C	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	2	2	2	1	2	1	2	1	2
CO 2	2	2	2	3	1	2	1	1	1	2
CO 3	2	3	2	3	2	2	1	2	1	3
CO 4	3	2	3	2	1	3	1	2	1	2

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level)

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Principles of Radio Broadcasting		15	1
	1.1	Radio often uses linear formats (e.g., news bulletins) but can include non-linear narrative styles in drama and podcasts., Uses and Gratifications Theory, Parasocial Interaction Theory, Hypodermic	10	

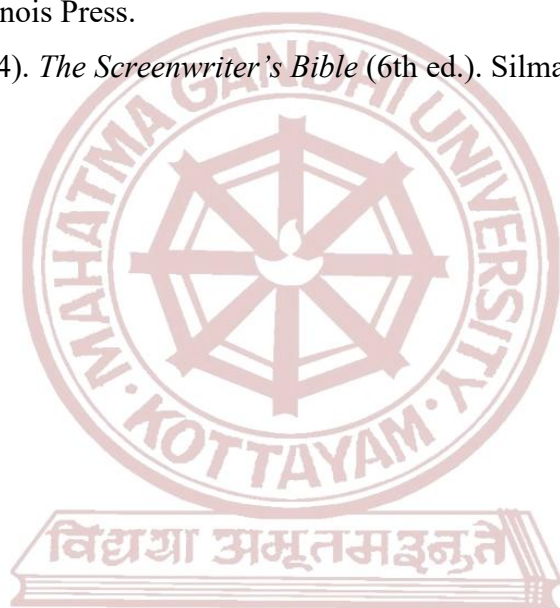
		Needle Theory (Magic Bullet Theory), Aural Semiotics		
	1.2	Radio Ethics and Regulations. Copyright laws. Broadcasting standards, Code of conduct for announcers	5	1
	Formats and Structure of Radio Programs		15	
2	2.1	The Structure of Radio Programs (Analyzes various radio formats like talk shows, features, news, etc, News Bulletins, Talk Shows, Radio Dramas, Music Programs, Documentary or Feature Programs, Public Service Announcements (PSAs), Phone-In Shows	9	2
	2.2	Common Radio Program Formats News Bulletins, Radio Dramas / Serials, Music Shows, Magazine Programs, Documentaries / Features, Phone-In Programs - Purpose, Features, Style, Example:	6	2
	Role of Script and Sound Design in Radio Production		15	
3	3.1	Role of Script and Sound Design in Radio Production (Explores the theory behind scripting, use of sound effects, and voice modulation.), Audience Psychology in Radio Programming (Discusses how radio programs are tailored based on listener behaviour and preferences.), Evolution and Impact of Radio Program Production Techniques (Historical and theoretical overview of how radio production has developed over time.)	8	3
	3.2	Elements of Effective Radio Content Design (Studies theoretical components like storytelling, flow, pacing, and engagement.), Communication Models in Radio Production (Examines how communication	6	3

		theories like Shannon-Weaver or Uses and Gratifications apply to radio		
4	4.1	Teacher Specific Content		

Teaching and Learning Approach	Interactive Theory Sessions • Mini-Lectures on each audio-communication theory (e.g. Uses & Gratifications, Aural Semiotics) illustrated with classic radio clips. • Guided Discussions to unpack how linear vs. non-linear storytelling affects listener engagement. Script & Sound Design Workshops • Hands-on scripting exercises: students draft short radio spots using proper format, voice cues, and pacing markers. • Sound-design practice: layering voice, music, and effects in simple audio editing software to reinforce flow and mood.
Assessment Types	MODE OF ASSESSMENT Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) Practical Total Marks: 25 FA: Quiz, Assignment, Oral Presentations SA: Practical test
	B. End Semester Evaluation (ESE) Practical Total Marks: 50, Duration: 3 hrs Project Evaluation: 40 marks Viva Voice: 10 marks

References

1. Horton, D., & Wohl, R. R. (1956). Mass communication and para-social interaction. *Psychiatry*, 19(3), 215–229.
2. Katz, E., Blumler, J. G., & Gurevitch, M. (1974). Uses and gratifications research. *The Public Opinion Quarterly*, 37(4), 509–523.
3. McLeish, R., & Link, J. (2015). *Radio Production* (6th ed.). Focal Press.
4. Rabiger, M., & Herman, M. (2020). *Directing the Documentary* (7th ed.). Routledge.
5. Shannon, C. E., & Weaver, W. (1949). *The Mathematical Theory of Communication*. University of Illinois Press.
6. Trottier, D. (2014). *The Screenwriter's Bible* (6th ed.). Silman-James Press.



MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering			
Programme	MGU-B.Voc. (Honours) Sound Engineering			
Course Name	Art of Critical Listening			
Type of Course	VAC			
Course Code	MG6VACSEG300			
Course Level	300			
Course Summary	This course develops critical listening skills essential for audio professionals. Students will learn to evaluate sound quality, understand listening environments, apply analytical listening techniques, and assess mixes across diverse genres. Topics include frequency balance, dynamics, stereo imaging, room acoustics, distortion detection, and reference-based listening to ensure accurate audio judgment across all playback systems.			
Semester	6	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
	Lecture class, Hands-on Practicum sessions	3	0	0
Pre- requisites, if any	Nil			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand audio quality by analyzing frequency balance, dynamic range, stereo imaging, and timbral characteristics.	U	1,2,3,4,6, 7,8,9,10

2	Optimize listening environments through understanding room acoustics, speaker placement, and calibration techniques.	A	1,2,3,4,5,6,7,8,9,10
3	Diagnose audio issues such as EQ imbalances, distortion, clipping, phase problems, and compression artifacts.	A	1,2,3,4,5,6,7,8,9,10
4	Evaluate mixes across genres and playback systems using reference tracks and comparative listening strategies.	E	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO1	3	3	2	1	0	1	1	1	1	2
CO2	2	3	2	1	1	1	1	1	1	2
CO3	2	3	2	2	1	1	1	1	2	2
CO4	2	3	2	1	2	1	1	1	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Frequency Analysis		15	
	1.1	Frequency and Dynamics – Explore the full frequency spectrum, achieve tonal balance, and manage dynamic range and headroom for clear, distortion-free audio.	6	1
	1.2	Stereo Imaging and Spatial Depth – Learn techniques to create a wide and immersive stereo field, with depth and dimension using panning, reverb, and delay	4	1
	1.3	Timbre and Harmonic Content – Understand the unique tonal qualities of sounds, analyze harmonic richness, and use tools to shape and enhance sonic character.	5	1

2	Listening Environments		15	
	2.1	Acoustic Fundamentals – Understand how room acoustics and monitoring conditions affect sound perception and accuracy.	5	2
	2.2	Monitoring Techniques – Compare nearfield and farfield listening setups and their impact on critical listening.	5	1
	2.3	Environment Optimization – Learn to calibrate listening spaces and recognize psychological factors that influence audio judgment.	5	2
3	Analytical Listening Techniques		15	
	3.1	Develop skills to identify EQ imbalances, distortion, clipping, phase issues, and compression artifacts. Use reference tracks to evaluate mix quality and consistency across playback systems.	5	3
	3.2	Analyse and assess mixes and masters across a range of genres, refining personal listening references for professional judgment.	10	4
4	4.1	Teacher Specific Content		

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) 1. Theory Classes for understanding frequency 2. Demonstrations of various sound aspects 3. Hearing practice sessions 4. Assessments: Regular quizzes, assignments, lab work, and practical exams to check understanding and skills.
Assessment Types	MODE OF ASSESSMENT Mode of Assessment

	A. Continuous Comprehensive Assessment (CCA) Practical Total Marks: 25 FA: Quiz, Assignment, Oral Presentations SA: Practical Test
	B. End Semester Evaluation (ESE) Practical Total Marks: 50, Duration: 2 hrs Lab Exam: 40 marks Viva Voice: 10 marks

References

1. Everest, F. Alton. *Critical Listening Skills for Audio Professionals*. Boston: Cengage Learning, 2007.
2. Izhaki, Roey. *Mixing Audio: Concepts, Practices and Tools*. 3rd ed. New York: Focal Press, 2018.
3. Rumsey, Francis, and Tim McCormick. *Sound and Recording: Applications and Theory*. 7th ed. New York: Focal Press, 2014.

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering				
Programme	MGU-B.Voc. (Honours) Sound Engineering				
Course Name	Project				
Type of Course	PRJ				
Course Code	MG6PRJSEG300				
Course Level	300				
Course Summary	The final project on 5.1 Surround Sound Production serves as a capstone experience for students of the Sound Engineering course, allowing them to apply their technical knowledge and creative skills in an immersive audio environment. This project emphasizes the theory and practical implementation of multichannel sound design, mixing, and spatial audio imaging. Students will engage in the full production workflow—from pre-production planning to surround mixing and final output—using industry-standard digital audio workstations and monitoring systems. The goal is to produce a professional-quality surround sound piece, such as a film scene, music track, or documentary segment that demonstrates an understanding of spatial audio aesthetics, technical accuracy, and effective storytelling through sound.				
Semester	6	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
	Practical sessions	0	8	0	
Pre- requisites, if any	Nil				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Design and plan a surround sound workflow for a music, film, or media production.	C	1,2,3,4,6,9,10

2	Apply advanced multichannel recording, panning, and mixing techniques using a 5.1 setup.	A	1,2,3,5,6,9,10
3	Troubleshoot and optimize spatial imaging, loudness, and signal flow issues in a surround session.	E	1,3,4,5,6,8,9,10
4	Create and present a completed 5.1 surround mix with documentation of tools, challenges, and solutions.	C	1,3,4,6,7,8,9,10

Assessment Types	MODE OF ASSESSMENT	
	Mode of Assessment	
	A. Continuous Comprehensive Assessment (CCA)	
	Total Marks: 30	
	Components	Marks
	Commitment and Involvement	5
	Periodic Progress Review	15
	Report	10
	B. End Semester Evaluation (ESE)	
	Total Marks: 70	
	Components	Marks
	Dialogue	10
	Sound Effects	10
	Final Output	40
	Viva Voice	10

SEMESTER 7 & 8



MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering				
Programme	MGU-B.Voc. (Honours) Sound Engineering				
Course Name	Apprenticeship				
Type of Course	APP				
Course Code	MG7APPSEG400				
Course Level	400				
Course Summary	The Apprenticeship in the B.Voc. (Honours) Sound Engineering programme is a structured experiential learning component conducted during the final semesters. It provides students with hands-on exposure in professional settings such as recording studios, broadcast stations, film post-production houses, and live sound environments. Under the mentorship of industry professionals, students actively engage in tasks like recording, mixing, sound design, and event coordination. The apprenticeship enhances technical proficiency, workplace discipline, communication skills, and industry readiness. Assessment is based on employer feedback, a detailed apprenticeship report, and a final review, ensuring reflective learning and real-world competence.				
Semester	7 & 8	Credits		28	Total Days
Course Details	Learning Approach	Lecture	Practical	OJT	
	Practical sessions	0	28	0	
Pre- requisites, if any	Nil				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Apply sound engineering knowledge in real-world project environments under professional guidance.	A	1,2,3,5,6,9,10

2	Demonstrate workplace discipline, studio etiquette, and collaborative production practices.	E	4,5,6,8,9,10
3	Operate industry-standard tools and workflows for recording, editing, mixing, or live sound operations.	S	1,2,3,5,6,9,10
4	Reflect on the apprenticeship experience and evaluate personal skill development and career readiness.	S	1,4,6,7,8,9,10

Assessment Types	MODE OF ASSESSMENT	
	Mode of Assessment	
	A. Continuous Comprehensive Assessment (CCA)	
	Total Marks: 90	
	Components	Marks
	Commitment, Punctuality & Professional Conduct	20
	Monthly Progress Review	25
	Skill Development & Application	25
	Report	20
	B. End Semester Evaluation (ESE)	
	Total Marks: 210	
	Components	Marks
	Feedback & Evaluation Report from Host Organization	50
	Skill Demonstration/Summary of the Work Exposure	40
	Final Report	40
	Domain Knowledge and Experience Communication (Presentation)	40
	Viva Voice	40



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Sound Engineering				
Programme	MGU-B.Voc. (Honours) Sound Engineering				
Course Name	Research Internship				
Type of Course	RIN				
Course Code	MG7RINSEG400				
Course Level	400				
Course Summary	The Research Internship course is intended to introduce students to academic and applied research within the domain of sound engineering. Conducted under faculty or institutional mentorship, students will undertake guided mini-projects, literature reviews, case studies, or exploratory investigations in areas such as audio technology, acoustic studies, psychoacoustics, sound design, or emerging media tools. The course fosters analytical thinking, research ethics, and technical writing skills while encouraging students to explore real-world problems and propose sound-related solutions. Assessment includes progress reports, a written research paper or project file, and a viva voce.				
Semester	7 & 8	Credits		20	Total Days
Course Details	Learning Approach	Lecture	Practical	OJT	
	Practical sessions	0	20	0	
Pre- requisites, if any	Nil				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Formulate a research question or hypothesis in the field of sound engineering.	An	1,2,3,4,10

2	Apply research methods such as literature review, data collection, and analysis.	A	1,2,3,5,6,10
3	Demonstrate awareness of research ethics, documentation, and referencing techniques.	E	3,4,6,7,8,10
4	Prepare and present a research report or mini-project based on findings and analysis.	C	1,3,4,6,8,9,10

Assessment Types	MODE OF ASSESSMENT	
	Mode of Assessment	
	A. Continuous Comprehensive Assessment (CCA)	
	Total Marks: 60	
	Components	Marks
	Commitment, Punctuality & Professional Conduct	10
	Monthly Progress Reviews	15
	Skill Development & Application	15
	Report	20
	B. End Semester Evaluation (ESE)	
	Total Marks: 140	
	Components	Marks
	Feedback & Evaluation Report from Host Organization	40
	Skill Demonstration/Summary of the Work Exposure	20
	Final Report	25
	Domain Knowledge and Experience Communication (Presentation)	25
	Viva Voice	30

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Syllabus